

Olist E-Commerce Platform: Concentration Risk & Revenue Recovery

Brazilian E-commerce Market Intelligence | 2016–2018

Author: Zachary Tanner

Date: 22 July 2026

Dataset: Brazilian E-commerce Public Dataset by Olist (Kaggle)

Pipeline: Databricks Medallion Architecture (Bronze → Silver → Gold)

Visualisation: Tableau Public

Executive Summary

The Olist platform processed 96,470 orders generating R\$15.42M in gross merchandise value between 2016 and 2018. This report presents a comprehensive analysis of platform performance across five analytical dimensions: sales and fulfilment, freight economics, seller health, regional disparities, and customer behaviour. The findings reveal a platform with strong operational fundamentals but material concentration risks and inherent retention vulnerabilities that, left unaddressed, represent significant revenue exposure.

Five findings demand immediate executive attention:

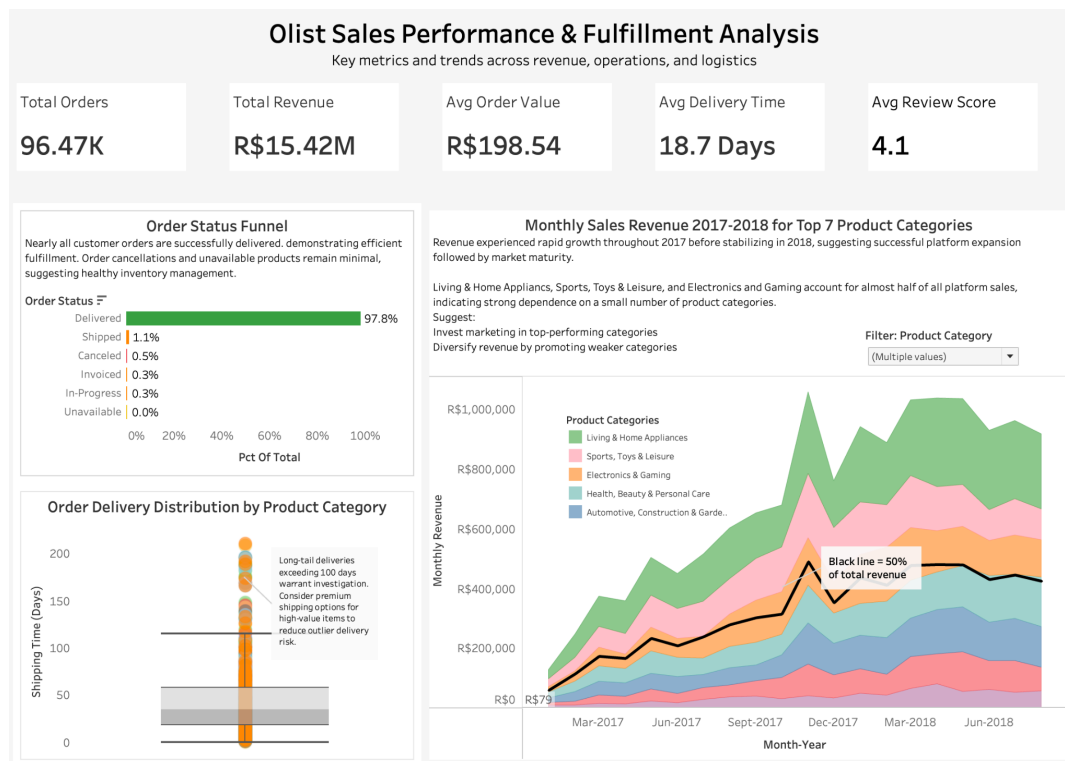
1. **Revenue concentration is dangerously narrow.** The top 136 sellers (representing only 4.4% of the active seller base) generate approximately 50% of total platform gross merchandise value (GMV). The top 30 sellers alone account for more than 25% of all revenue. This level of dependency creates a single point of failure: the loss of even a small number of top sellers would have an immediate and measurable impact on platform sustainability. Retention strategy for this cohort is the platform's most material financial risk.
2. **The platform's highest-revenue seller tier is also its most operationally fragile.** High Revenue / Poor Delivery sellers account for 48% of total GMV (R\$7.45M) yet average 13.9 days to delivery. This is 27% slower than the Star Sellers tier. Customer satisfaction within the High Revenue / Poor Delivery tier (4.0), while slightly below the platform average (4.1), is a lagging indicator. Sustained delivery underperformance will likely erode satisfaction scores over time. Proactive service level agreement (SLA) engagement with this cohort is a high-leverage operational intervention available to the platform.
3. **Freight economics threaten margin sustainability in key categories.** 904 products carry freight ratios exceeding the 30% pipeline risk threshold, representing R\$540,984 in revenue at risk from margin erosion. 24 products in particular have been flagged for both surpassing this threshold and demonstrating multiple 3-month cycles of declines exceeding 40%. These products should be prioritized for review as discontinuation candidates. Low price, high freight expense products cannot absorb shipping costs through margin and are declining in demand simultaneously: a dual signal for urgent delisting or repricing review.

4. **R\$1.1M in high-value customer revenue sits with dissatisfied buyers.** Of the platform's 95,026 unique customers, 1,226 high-value customers (those with lifetime spend above R\$500) carry an average review score below 4.0. Their average spend of R\$931 per customer is nearly 7 times higher than the low-value segment. These customers have demonstrated platform trust through their spending but have expressed disappointment in their experience. Individualised recovery outreach for this cohort is economically justified and analytically tractable.
5. **Geographic revenue concentration mirrors seller concentration.** The Southeast region generates 64.6% of total platform revenue, with São Paulo alone accounting for R\$5.77M (37.4%). Northern and Northeastern states are systematically underserved. Lower seller density, longer delivery times, and below-average on-time delivery rates compound into a structurally inferior customer experience that suppresses demand growth in these regions. This gap could be addressed through targeted seller recruitment.

Section 1: Platform Overview and Sales Performance

1.1 Platform Scale and Growth

The Olist marketplace operates as an aggregation platform connecting independent Brazilian sellers with customers across all 26 states and the Federal District. Over the observation period, the platform processed 96,470 orders from 95,026 unique customers through 3,095 active sellers, spanning 32,951 unique products across 10 consolidated product categories.



Revenue grew rapidly throughout 2017. This was driven by expanding seller onboarding and growing consumer awareness of the marketplace model. Growth began to plateau in early 2018, with monthly revenue stabilising in the R\$800K–R\$1.1M range through mid-2018. This stabilisation pattern is consistent with market maturation rather than demand contraction. Active seller counts continued rising through the same period, suggesting the platform was successfully building supply capacity in anticipation of growing demand.

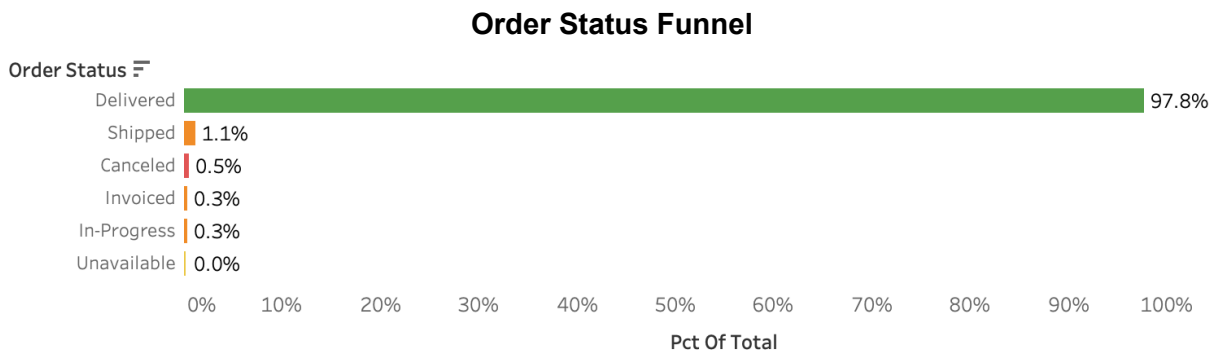
1.2 Key Performance Indicators

Metric	Value	Context
Total Orders	96,470	2016-2018
Total Revenue (GMV)	R\$15.42M	Gross Merchandise Value
Average Order Value	R\$198.54	Across all delivered orders
Average Delivery Time	18.7 days*	Order placement to customer receipt
Average Shipping Time	12.5 days*	Carrier pickup to customer receipt
Average Review Score	4.1 / 5	Across all reviewed orders
Delivery Success Rate	97.8%	Orders successfully delivered

*The 6.2-day gap between average shipping time and average delivery time reflects the processing and approval period between order placement and carrier handoff.

1.3 Order Fulfilment Pipeline

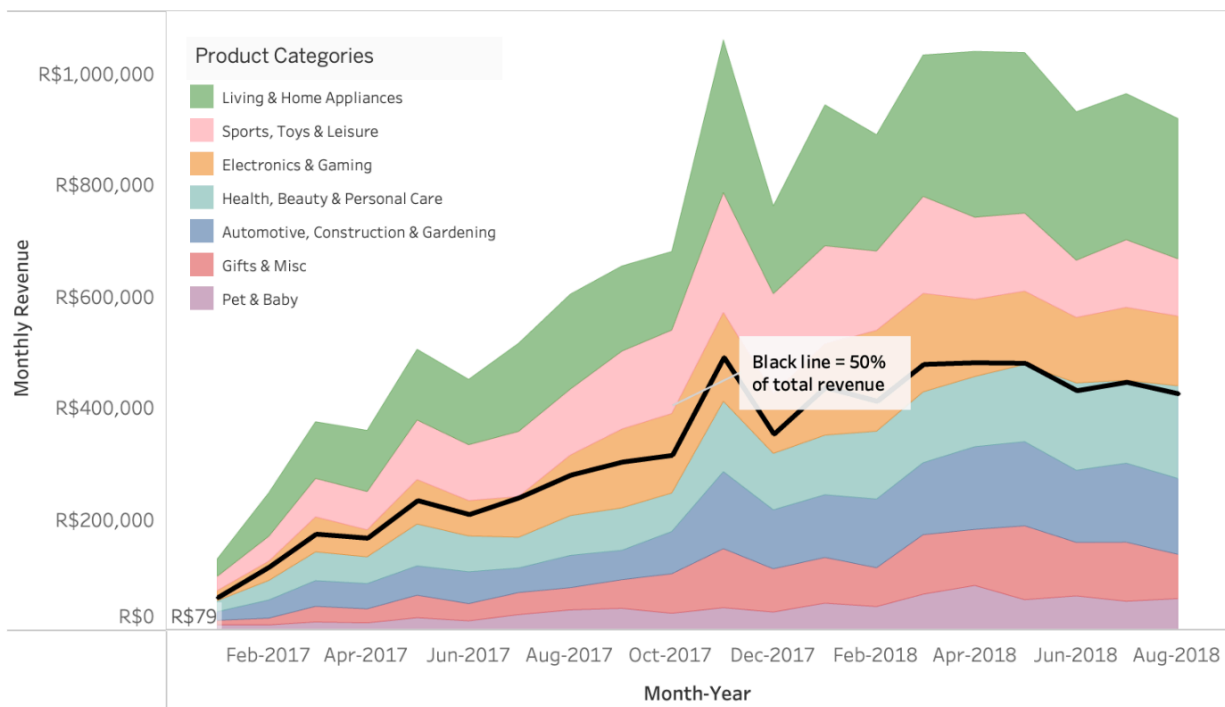
The platform's fulfilment pipeline demonstrates exceptional efficiency at the macro level: 97.8% of orders reach successful delivery status. Cancellation rates (0.5%) and unavailability rates are negligible, indicating healthy inventory management and order processing discipline.



One notable nuance in the cancellation data is that 776 orders exist in the platform's order system with no corresponding item or payment records. These represent orders cancelled before any transaction processing occurred. These have been excluded from revenue analysis but are retained in operational reporting as a signal of pre-purchase abandonment behaviour.

1.4 Category Revenue Concentration

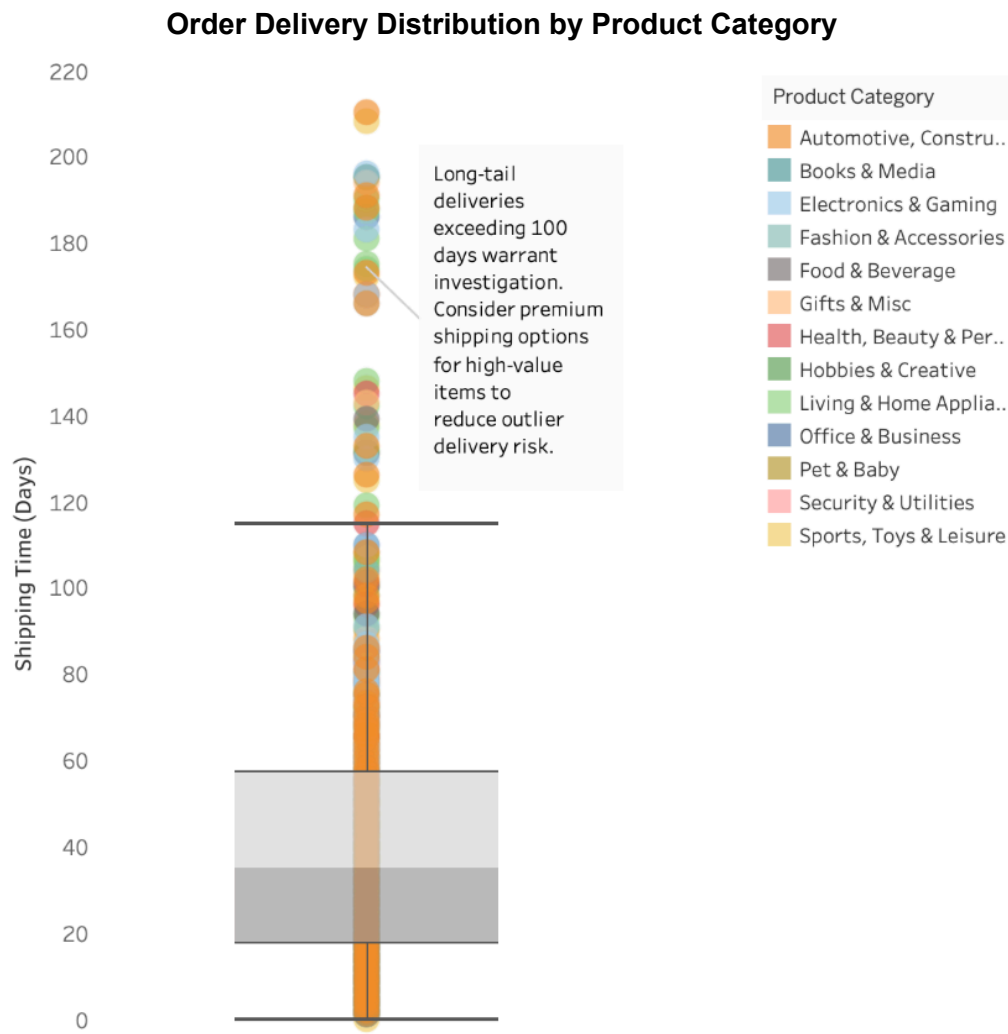
Monthly Sales Revenue 2017-2018 for Top 7 Product Categories



Three product categories (Living & Home Appliances, Sports, Toys & Leisure, and Electronics & Gaming) collectively account for approximately 50% of total platform revenue. This concentration creates both an opportunity (these categories have proven product-market fit and can absorb increased marketing investment) and a risk (platform revenue is disproportionately sensitive to demand shocks in these categories). Health, Beauty & Personal Care showed the most consistent growth trajectory throughout the observation period, suggesting a category still in its platform growth phase with headroom for continued expansion.

Strategic implication: The category mix suggests Olist's customer base skews toward considered, higher-ticket purchases (appliances, electronics) rather than low-cost, fast-moving consumables. Marketing investment in top-performing categories reinforces existing customer behaviour; a parallel strategy to promote underperforming categories diversifies revenue concentration risk.

1.5 Delivery Distribution by Category



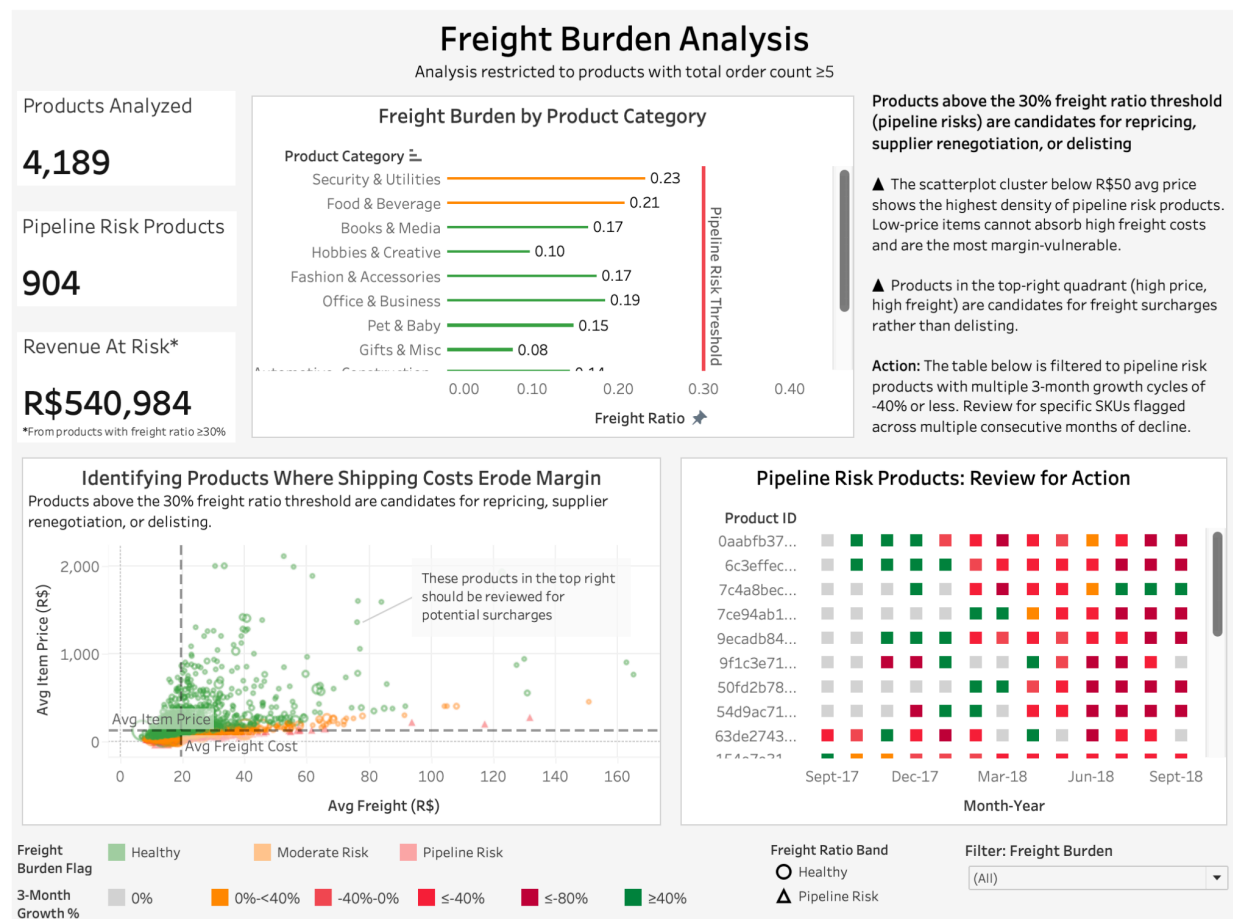
Delivery time distribution varies significantly across product categories. The box plot reveals that the upper whisker representing the slowest 25% of deliveries extends dramatically in categories with heavy or bulky products. Long-tail deliveries exceeding 100 days were identified across multiple categories. These represent both a customer satisfaction risk and a potential indicator of fulfillment bottlenecks at the carrier or seller level.

The presence of extreme outliers suggests that premium or expedited shipping options for high-value items would disproportionately improve satisfaction scores for a small but commercially important segment of orders.

Section 2: Freight Economics and Margin Analysis

2.1 The Freight Burden Challenge

Freight cost management is the most direct lever available for improving platform-wide margin sustainability. In a marketplace model where the platform facilitates rather than controls fulfilment, freight cost is a variable that compounds across every transaction, and this disproportionately impacts low-price, high-volume product categories.

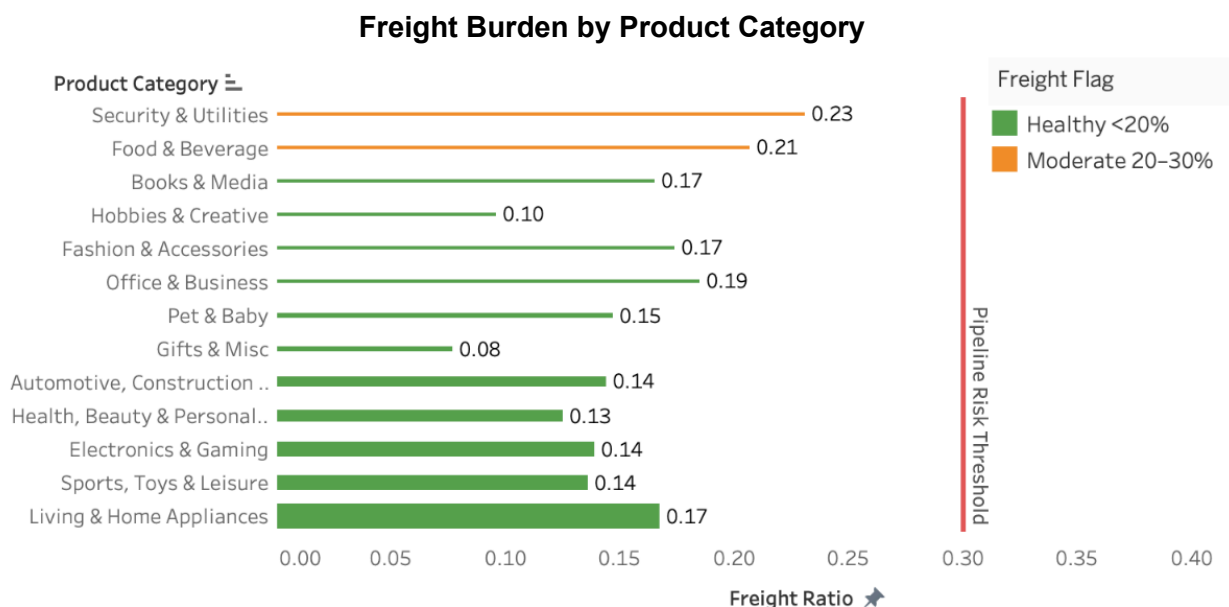


Analysis of freight ratios (freight cost as a proportion of total item value) across 4,189 products with a total order count of at least 5 identified three distinct risk tiers detailed in the next table:

Tier	Freight Ratio	Products	Description
Healthy	<20%	2,151	Freight cost is commercially sustainable
Moderate	20-30%	1,134	Approaching threshold, monitor closely
Pipeline Risk	$\geq 30\%$	904	Freight cost materially erodes margin

904 products (21.6% of the analysed product universe) carry freight ratios at or above the 30% pipeline risk threshold, representing R\$540,984 in revenue exposed to margin compression.

2.2 Category-Level Freight Burden



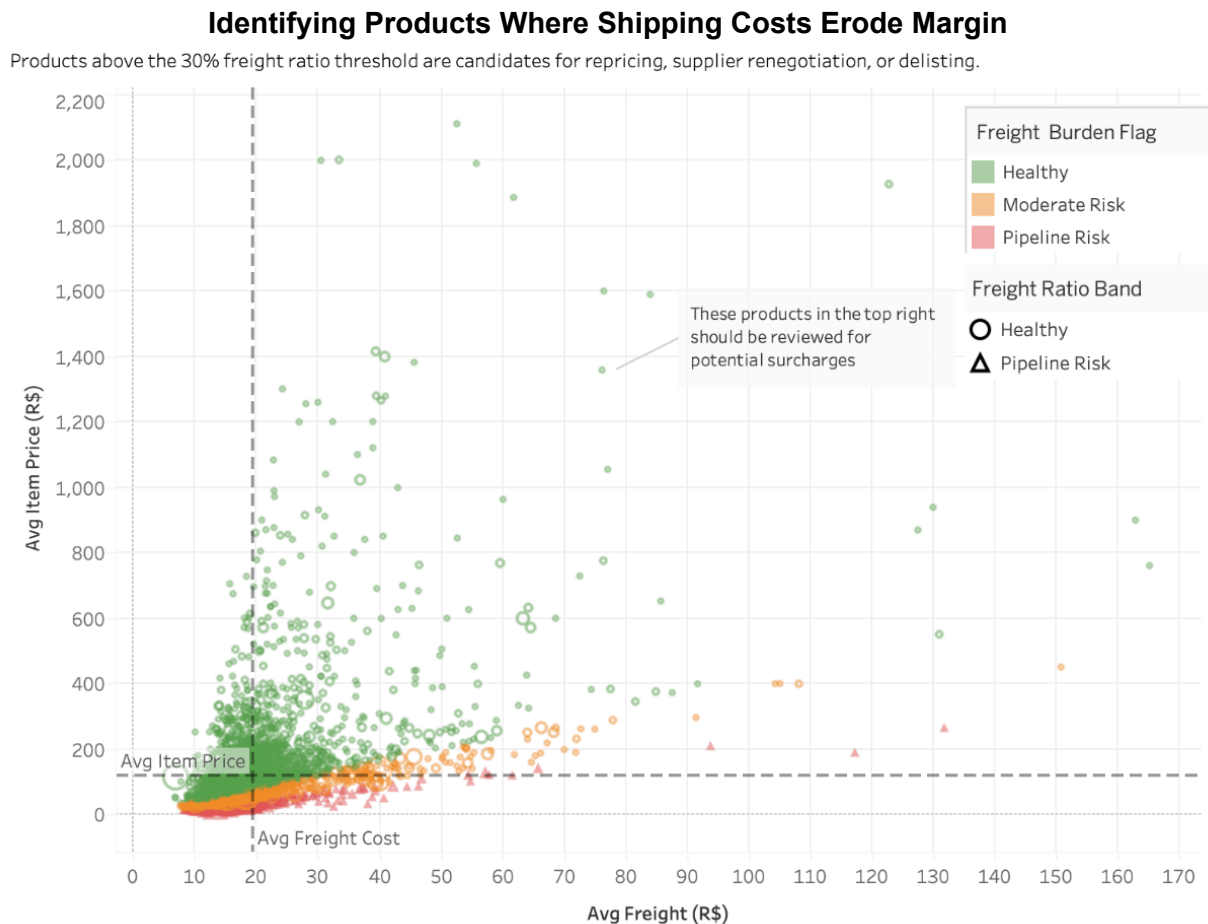
Security & Utilities carries the highest freight ratio at 0.23, followed by Food & Beverage at 0.21. No category currently exceeds the threshold at the aggregate level, but this masks significant product-level variation within categories. The pipeline risk product table reveals individual SKUs within otherwise healthy categories that have crossed the threshold and demonstrated multiple cycles of negative 3-month growth ([see Section 2.4](#)).

This freight burden distribution has direct implications for category strategy:

Security & Utilities and Food & Beverage: These categories combine relatively low-to-average item prices with products that are heavy or bulky relative to their value. The economic logic of selling these products through a marketplace with consumer-paid freight is inherently challenged. Supplier negotiation on product pricing, bundle strategies to increase average order value, or freight subsidisation programmes could be explored.

Health, Beauty & Personal Care: This category represents the lowest freight ratio in the analysed set. Light products with high value density make this category the most freight-efficient across the platform. This factor partially explains its consistent revenue growth trajectory.

2.3 The Price-Freight Relationship



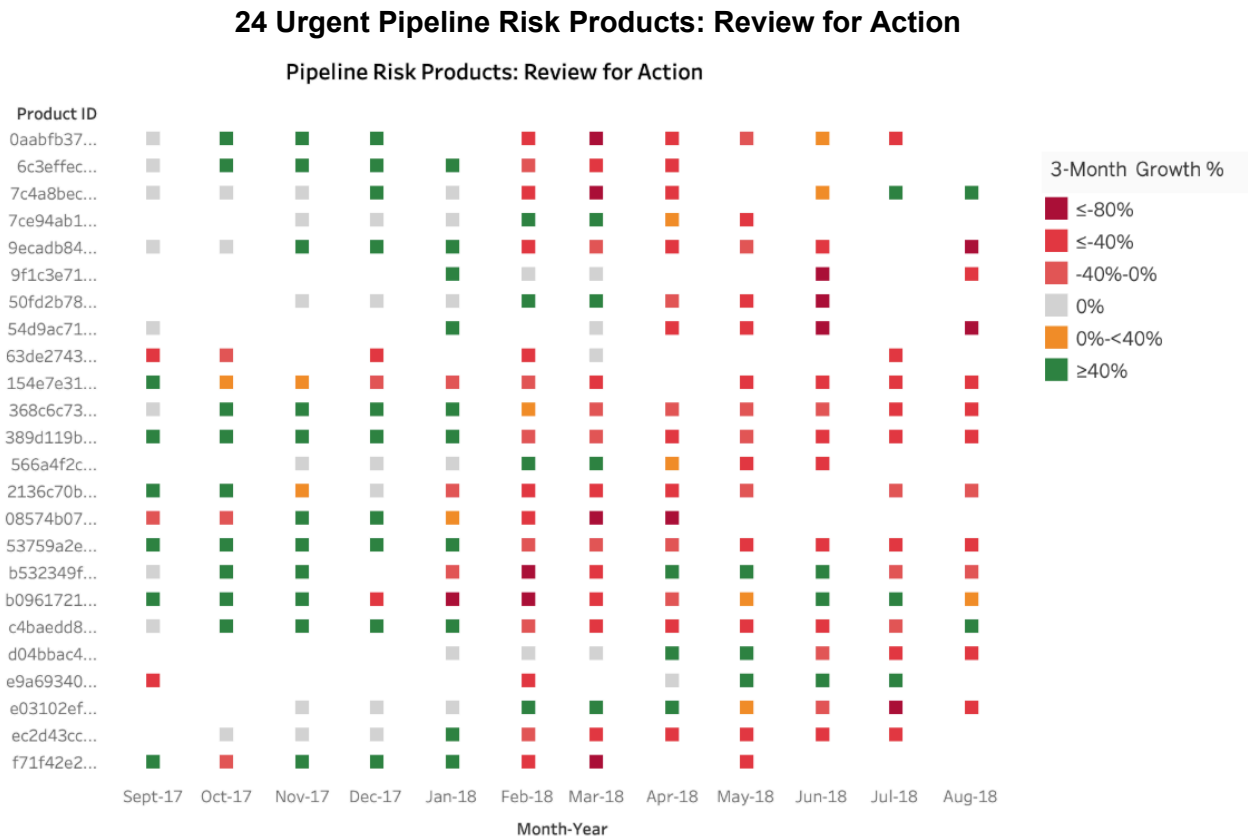
The scatterplot of average item price against average freight cost reveals a clear structural pattern: products below R\$50 in average selling price face the most severe freight burden, as fixed shipping costs consume a disproportionate share of product value. This cluster, visible as a dense concentration in the bottom-left quadrant of the scatter, represents the highest density of pipeline risk products.

Four distinct strategic zones emerge from the price-freight quadrant analysis:

1. **Top-left (High Price, Low Freight):** Premium margin products. This is the platform's most commercially attractive segment. Growth investment here offers the highest return value. Health, Beauty & Personal Care products dominate this quadrant.
2. **Top-right (High Price, High Freight):** Products where the margin exists to absorb freight costs but where a freight surcharge or premium delivery tier would be commercially viable. Candidates for repricing rather than delisting.
3. **Bottom-left (Low Price, Low Freight):** Volume-driven categories with thin but sustainable margins. Platform growth in these categories depends on order frequency more than order value.

- Bottom-right (Low Price, High Freight):** The margin squeeze zone. These products cannot absorb freight costs through their selling price. Products here are simultaneously experiencing demand decline (as identified in the rolling growth analysis) and represent the clearest candidates for platform delisting.

2.4 Pipeline Risk Product Identification



The 3-month growth analysis identified products satisfying three simultaneous criteria:

- Freight ratio $\geq 30\%$
- Rolling 3-month demand decline of under -40% for multiple cycles
- A minimum combined volume of 10 purchased units across the measurement window

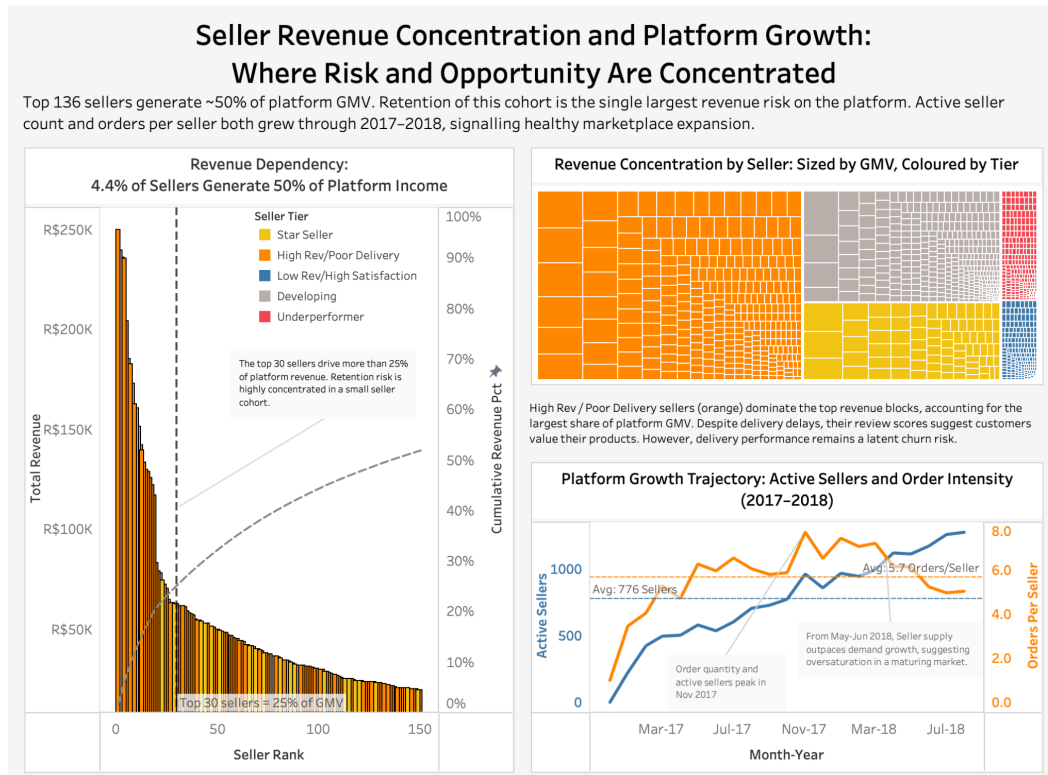
The 24 products flagged in this table represent the platform's most commercially urgent freight burden cases. These are products where declining demand and high freight costs are reinforcing each other in a negative feedback loop. These items should be prioritised for immediate commercial review and evaluated as candidates for delisting.

Addressing this freight burden through targeted delisting, repricing, or supplier renegotiation represents the platform's most direct and immediately actionable lever for margin recovery, and is prioritised accordingly in [Section 7: Strategic Recommendations](#).

Section 3: Seller Performance Analysis

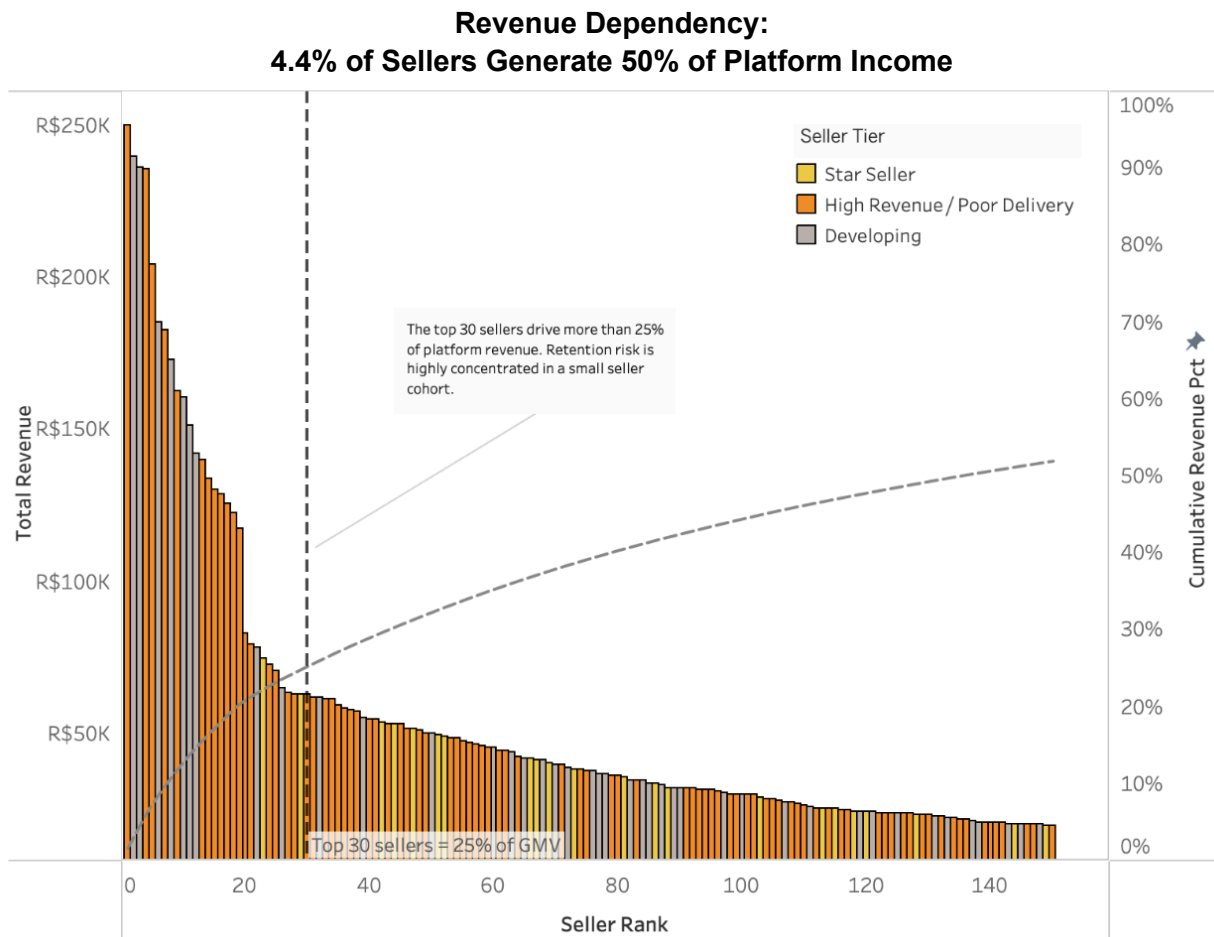
3.1 Seller Ecosystem Overview

The Olist seller ecosystem comprises 3,095 active sellers (defined as those with 10 or more fulfilled orders during the observation period). These sellers are distributed across all five Brazilian macro-regions, with a significant concentration in the Southeast. This mirrors the customer distribution pattern discussed in [Section 4](#).



Seller performance was evaluated across three primary dimensions: revenue contribution, delivery performance, and customer satisfaction. Sellers were assigned to one of five tiers based on NTILE quartile scoring across these dimensions.

3.2 Revenue Concentration: The Pareto Reality

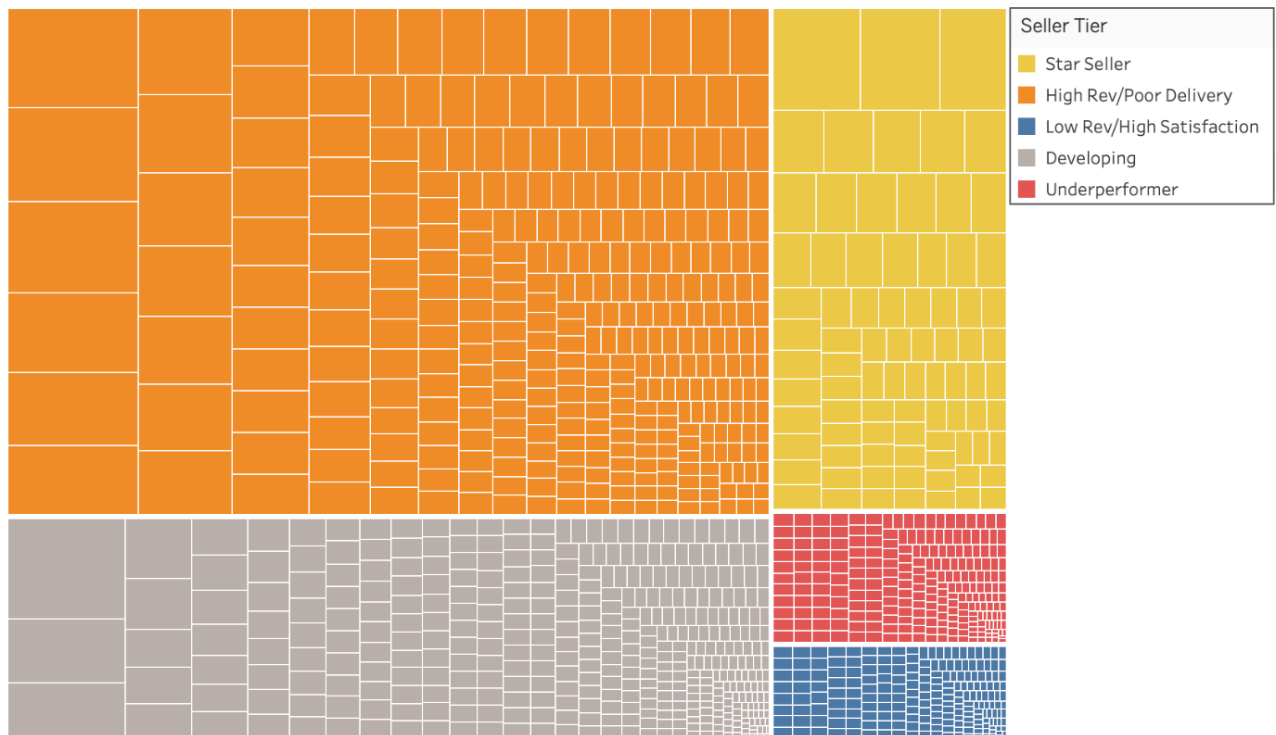


The platform's revenue distribution follows an extreme Pareto pattern:

- The **top 30 sellers** (< 1.0% of the seller base) generate more than 25% of total GMV
- The **top 136 sellers** (4.4% of the seller base) generate approximately 50% of total GMV
- The remaining **2,957 sellers** (95.6% of the seller base) collectively generate the remaining 50%

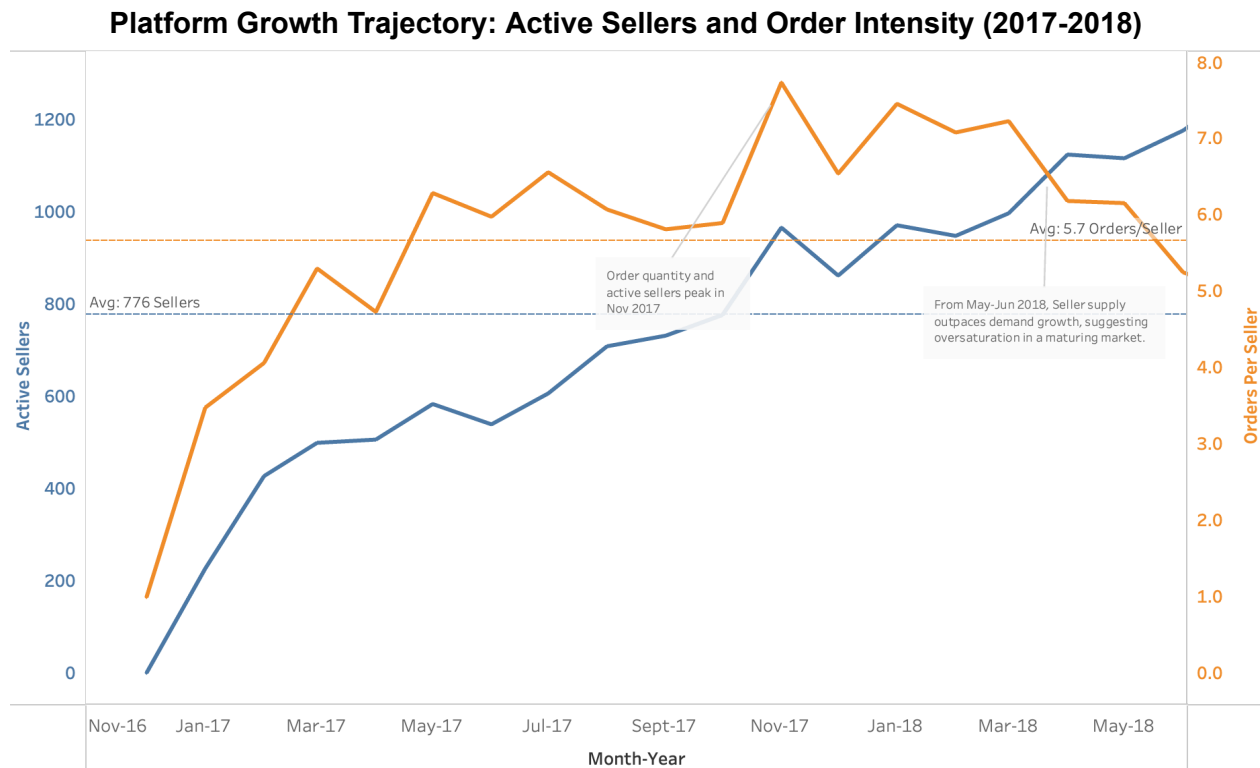
This concentration creates considerable retention risk. A churn event involving just 10 to 15 top-ranked sellers would be immediately visible in platform revenue. This balance is not inherently problematic but it demands a deliberate and resourced seller retention programme focused on this cohort.

Revenue Concentration by Seller: Sized by GMV, Coloured by Seller Tier



The treemap visualisation makes this seller concentration visceral: High Revenue / Poor Delivery sellers (orange) occupy the largest blocks in the platform's revenue landscape. Their commercial dominance is clear; their operational vulnerability is equally clear from the scorecard analysis in [Section 3.4](#).

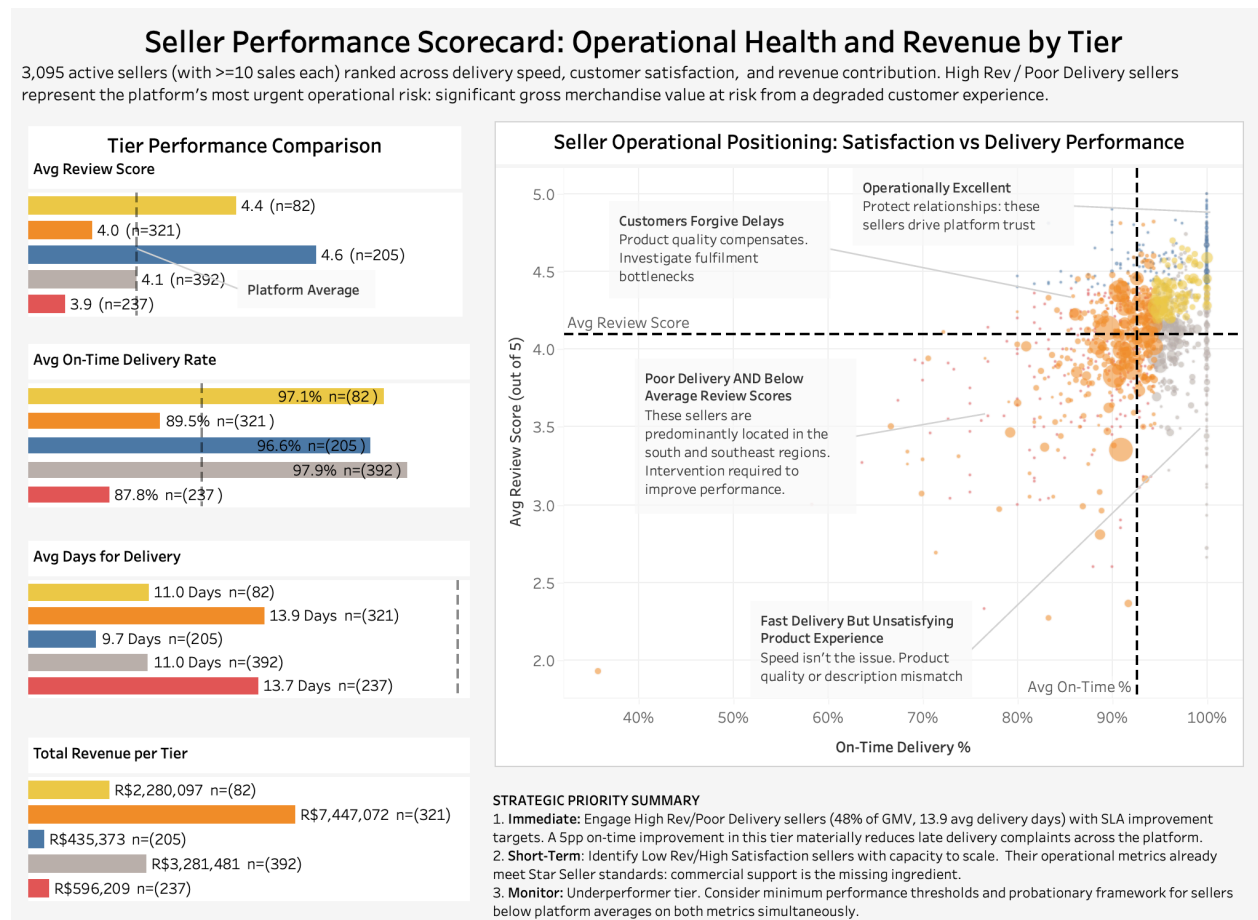
3.3 Platform Growth Trajectory



The active seller count grew from near-zero in late 2016 to over 1,000 by mid-2018, demonstrating substantial marketplace supply expansion. Orders per seller peaked at 7.7 in November 2017 before declining through 2018 as new seller onboarding outpaced demand growth.

This divergence between seller supply and orders per seller from May–June 2018 onwards is consistent with market saturation dynamics. In a maturing marketplace, seller productivity (orders per seller) is the more reliable health metric than raw seller count. The declining orders-per-seller trend warrants monitoring. If it continues, new seller onboarding should be paired with demand generation investment to better restore the supply-demand balance.

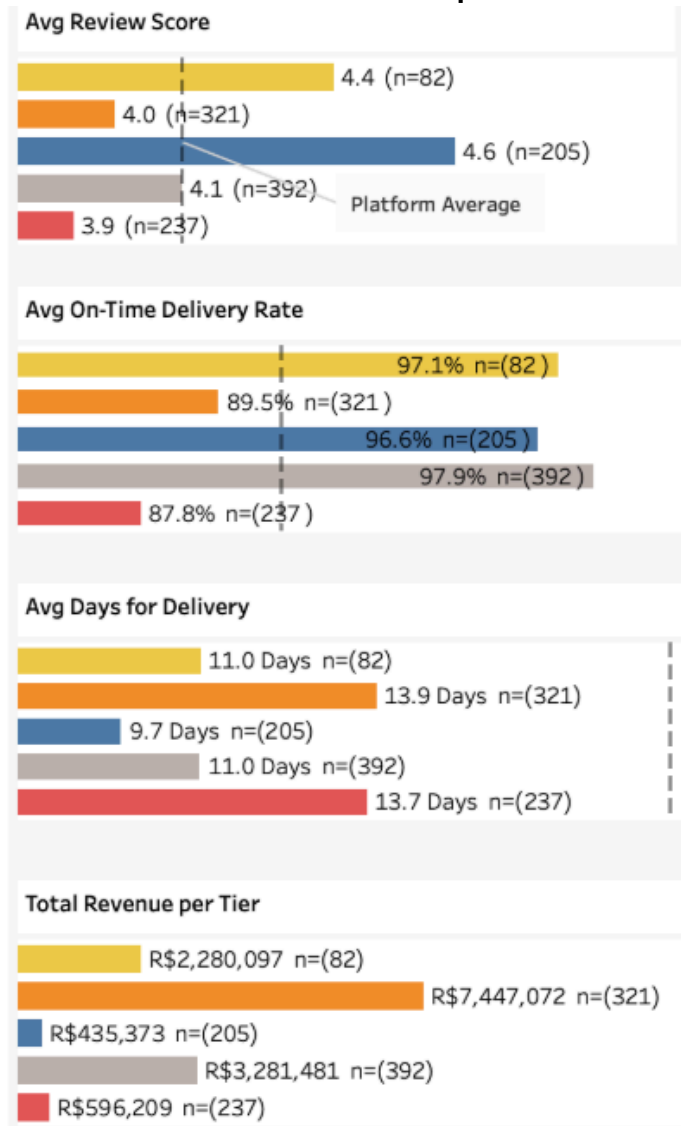
3.4 Seller Performance Scorecard



Sellers were scored across three quartile dimensions: revenue, on-time delivery, and review score and then assigned to one of five performance tiers. The tier distribution and key metrics are summarised below:

Tier	Sellers (n)	Total Revenue	Avg On-Time Delivery %	Avg Review Score	Avg Delivery Days
Star Seller	82	R\$2,280,097	97.1%	4.4	11.0
High Rev / Poor Delivery	321	R\$7,447,072	89.5%	4.0	13.9
Low Rev / High Satisfaction	205	R\$435,373	96.6%	4.6	9.7
Developing	392	R\$3,281,481	97.9%	4.1	11.0
Underperformer	237	R\$596,209	87.8%	3.9	13.7

Tier Performance Comparison



High Revenue / Poor Delivery is the tier that demands the most nuanced strategic response. These 321 sellers generated R\$7.45M (48% of total platform GMV) yet averaged 13.9 delivery days, 27% slower than Star Sellers. Their review scores (4.0 average) currently sit at the platform average, suggesting customers are tolerating delivery delays because the product quality compensates for the wait time. This tolerance is not unlimited. As customer expectations for delivery speed continue to rise, this tier's review scores will come under pressure unless delivery performance improves. The platform's optimal intervention is proactive SLA engagement, thereby establishing performance targets and providing logistical support before satisfaction scores deteriorate and endanger the commercial relationship.

Seller Operational Positioning: Satisfaction vs. Delivery Performance



Low Revenue / High Satisfaction sellers represent the platform's most underutilised growth asset. With average review scores of 4.6 (highest of any tier) and on-time delivery rates of 96.6%, these 205 sellers already operate on Star Seller operational standards. Further investigation of this tier's comparatively lower revenue is warranted. It may be attributable to catalogue depth and marketing visibility rather than operational capability. Commercial support, catalogue expansion incentives, and marketing co-investment could boost a meaningful proportion of this tier toward Star Seller status.

Underperformer sellers score below platform average on both delivery and satisfaction. With 237 sellers generating R\$596K, the individual commercial stakes are low. A minimum performance threshold framework with a defined improvement period and consequences for non-compliance would address this tail without significant revenue risk.

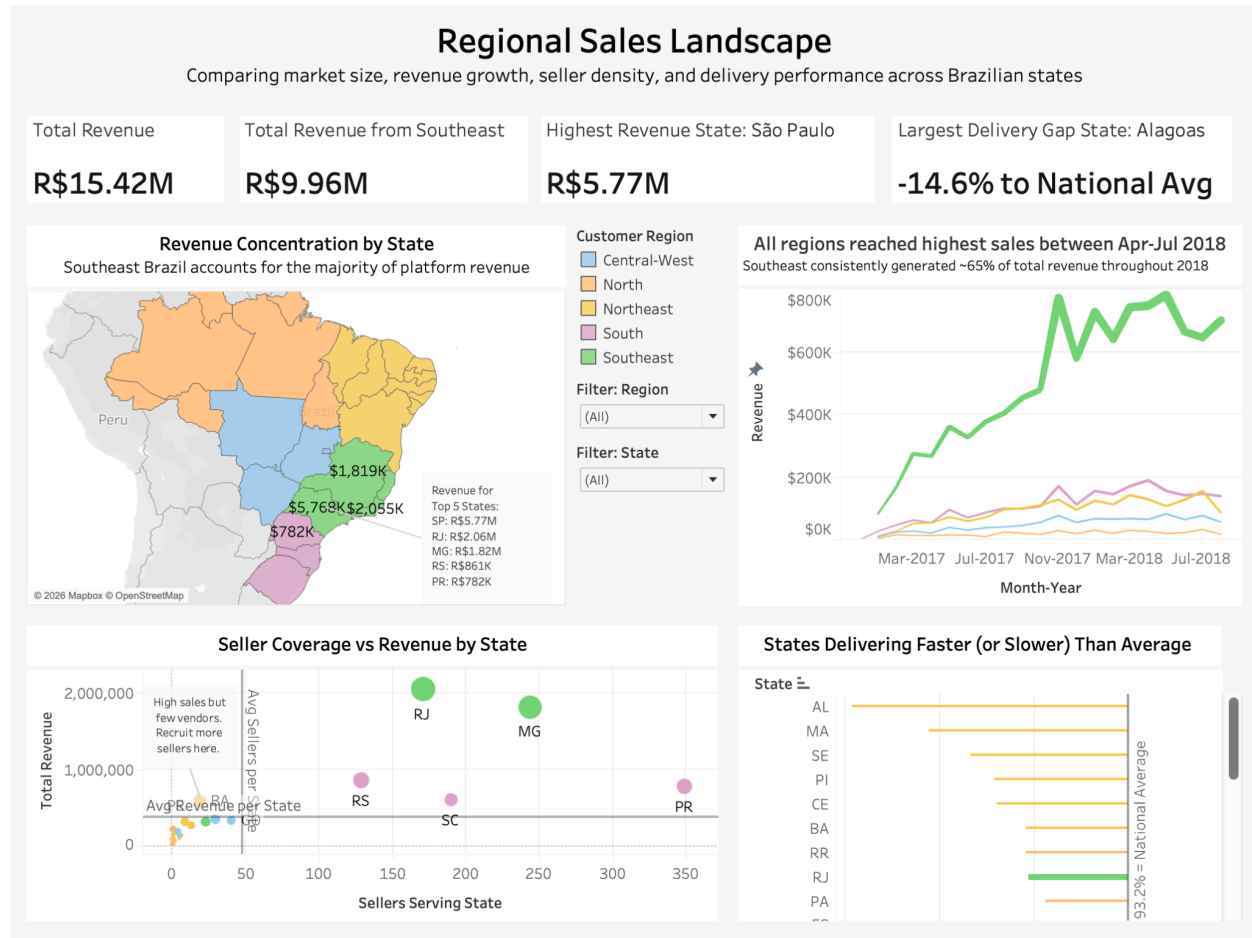
3.5 Geographic Seller Distribution

Seller recruitment is heavily concentrated in the Southeast, with São Paulo state alone hosting the largest share of active sellers. This geographic concentration has direct implications for delivery performance in the underserved Northern and Northeastern regions. A regionally targeted seller recruitment programme in the Northeast and North would simultaneously address delivery performance gaps and foster demand in markets where Olist currently has sub-optimal customer experience.

Taken together, the concentration risk in the top seller cohort, the operational fragility of the High Revenue/Poor Delivery tier, and the geographic imbalance in seller coverage point to the same conclusion: seller-side intervention is the platform's highest-leverage lever for both revenue protection and growth, and forms the basis of three of the six initiatives outlined in [Section 7's](#) priority matrix.

Section 4: Regional Analysis

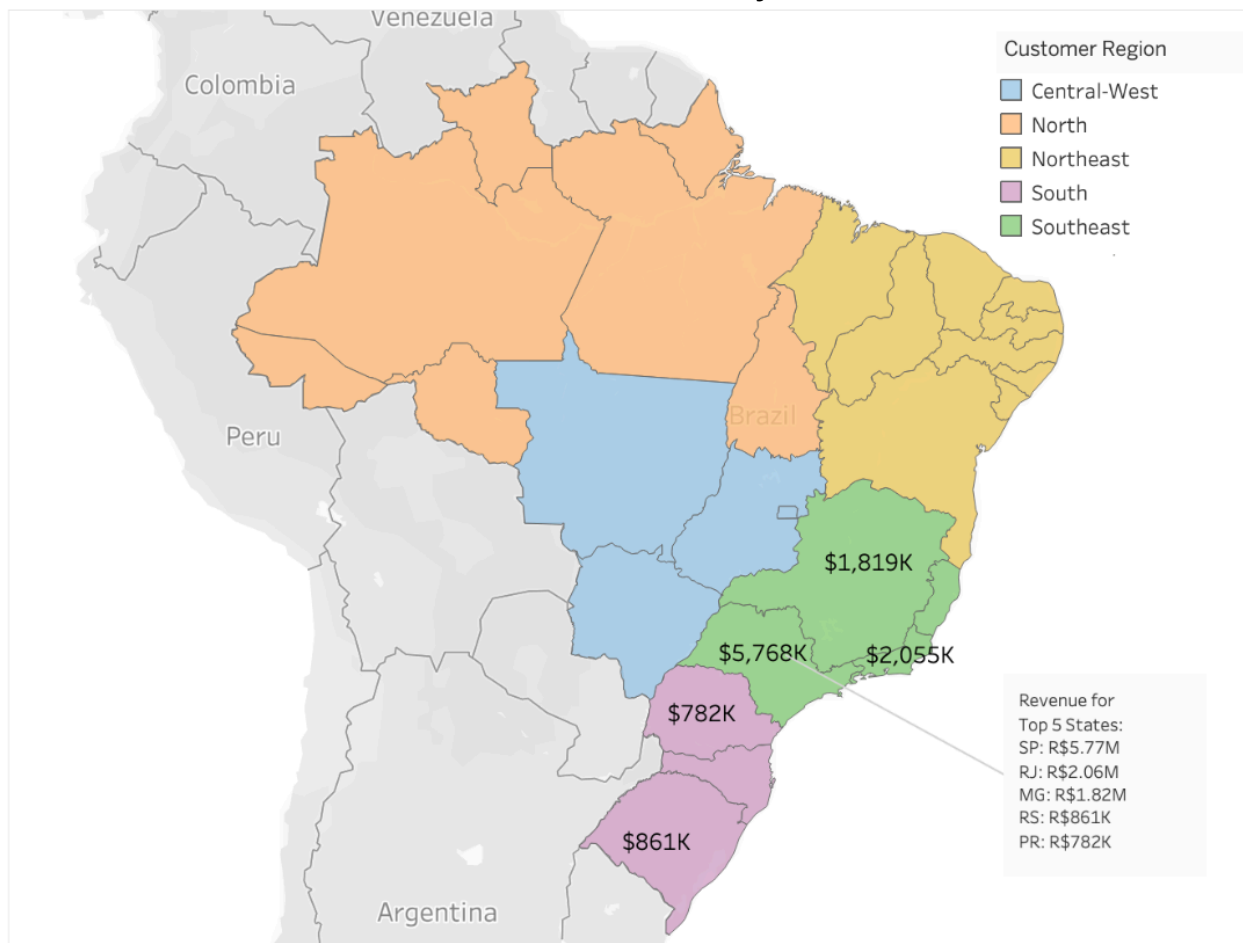
4.1 Geographic Revenue Distribution



Brazil's geographic scale and regional economic diversity create a highly uneven marketplace distribution. Platform revenue is concentrated in the Southeast to a degree that exceeds even the underlying population distribution.

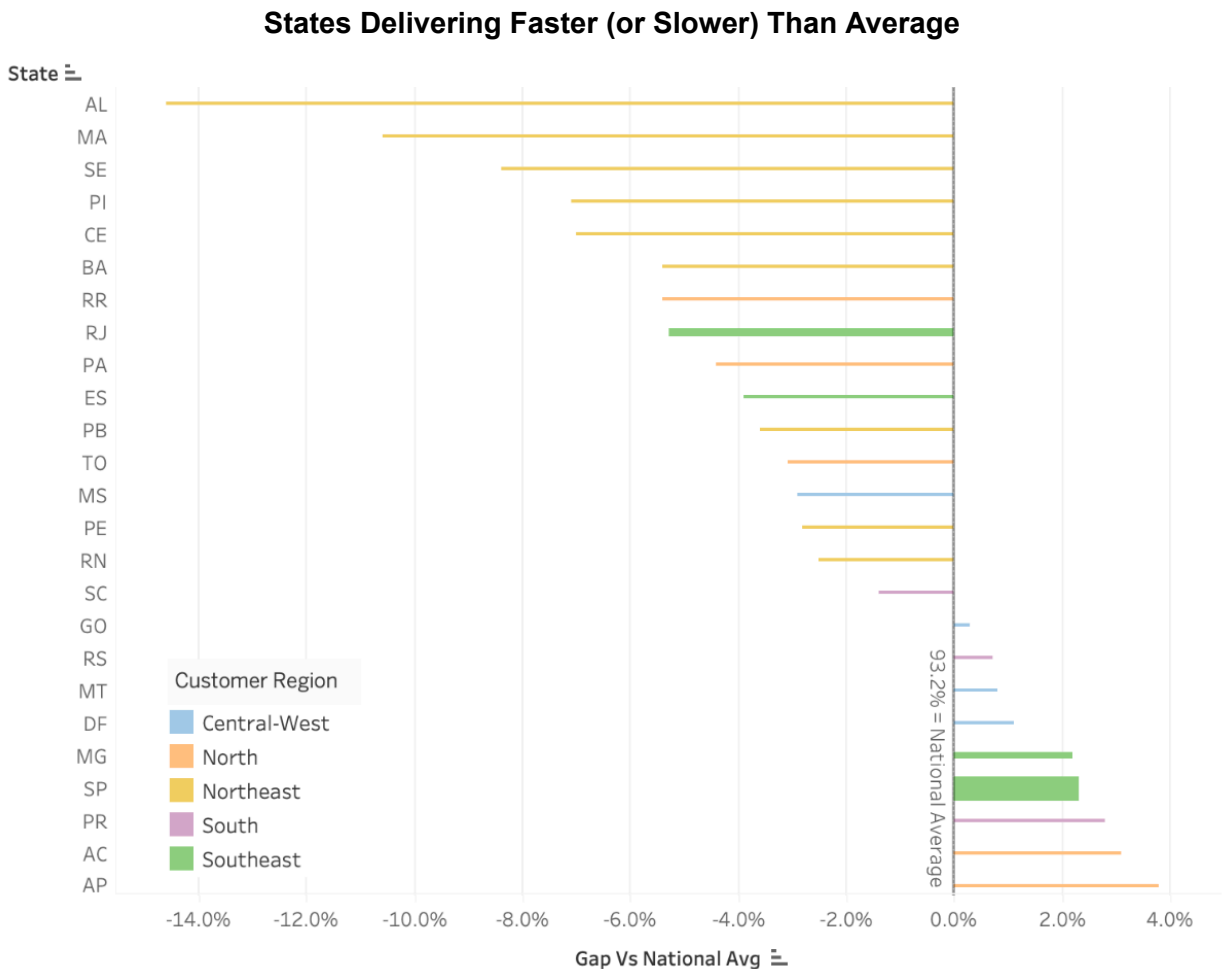
Region	Revenue	Share	Avg On-Time Delivery %
Southeast	R\$9.96M	64.6%	92.0%
South	R\$2.24M	14.5%	93.9%
Northeast	R\$1.82M	11.8%	86.3%
Central-West	R\$996K	6.5%	93.0%
North	R\$401K	2.6%	93.5%

Revenue Concentration by State



São Paulo state (R\$5.77M, 37.4% of total revenue) is the platform's dominant market by a wide margin. Rio de Janeiro (R\$2.06M) and Minas Gerais (R\$1.82M) round out the top three, together with São Paulo accounting for approximately 62% of total platform revenue.

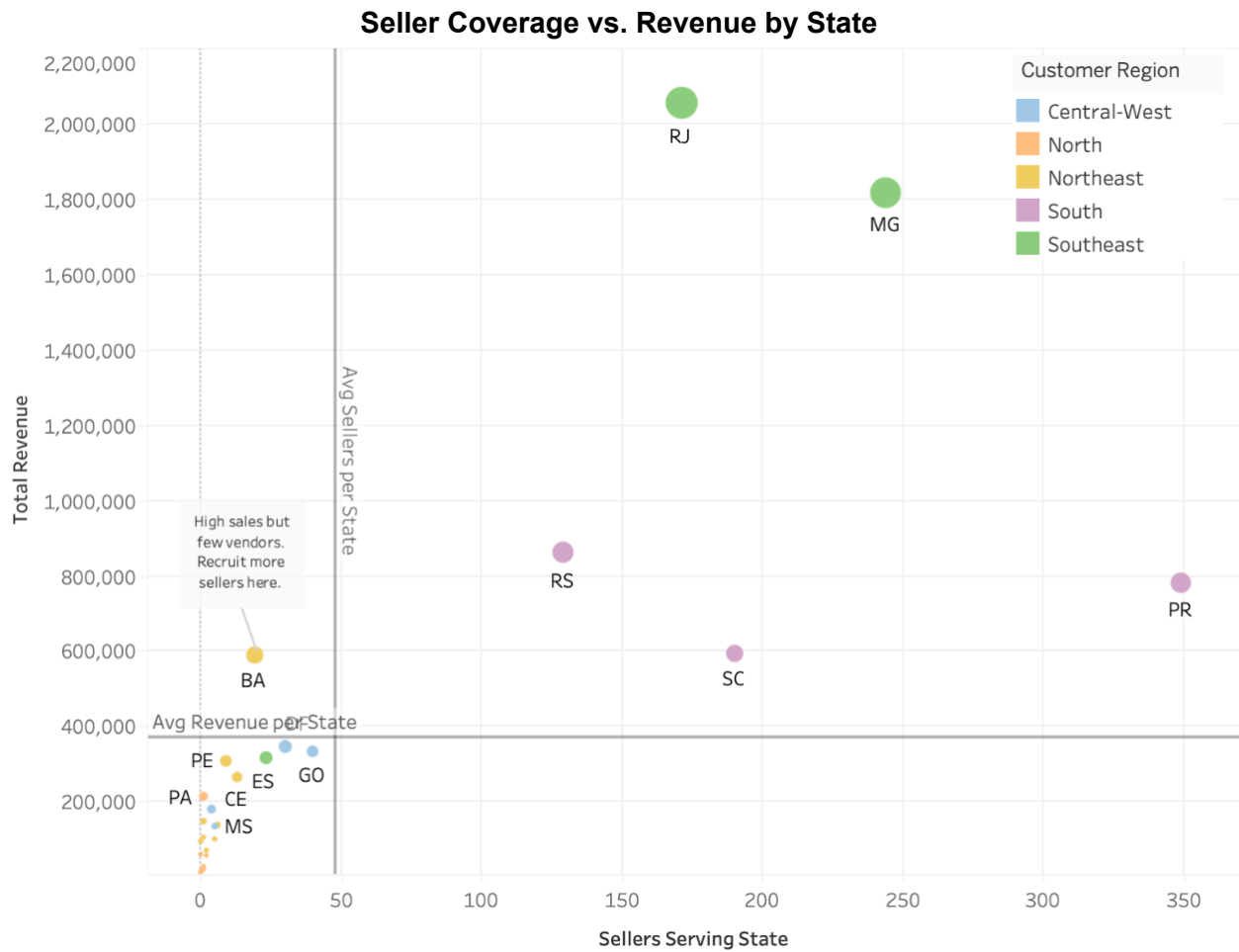
4.2 The Delivery Performance Gap



On-time delivery performance varies dramatically by state. Alagoas represents the most extreme underperformance at 14.6% below the national average of 93.2%. Northern states (Amazonas, Roraima, Amapá) consistently cluster at the bottom of the delivery performance ranking, reflecting geographic distance from seller concentrations and infrastructure challenges.

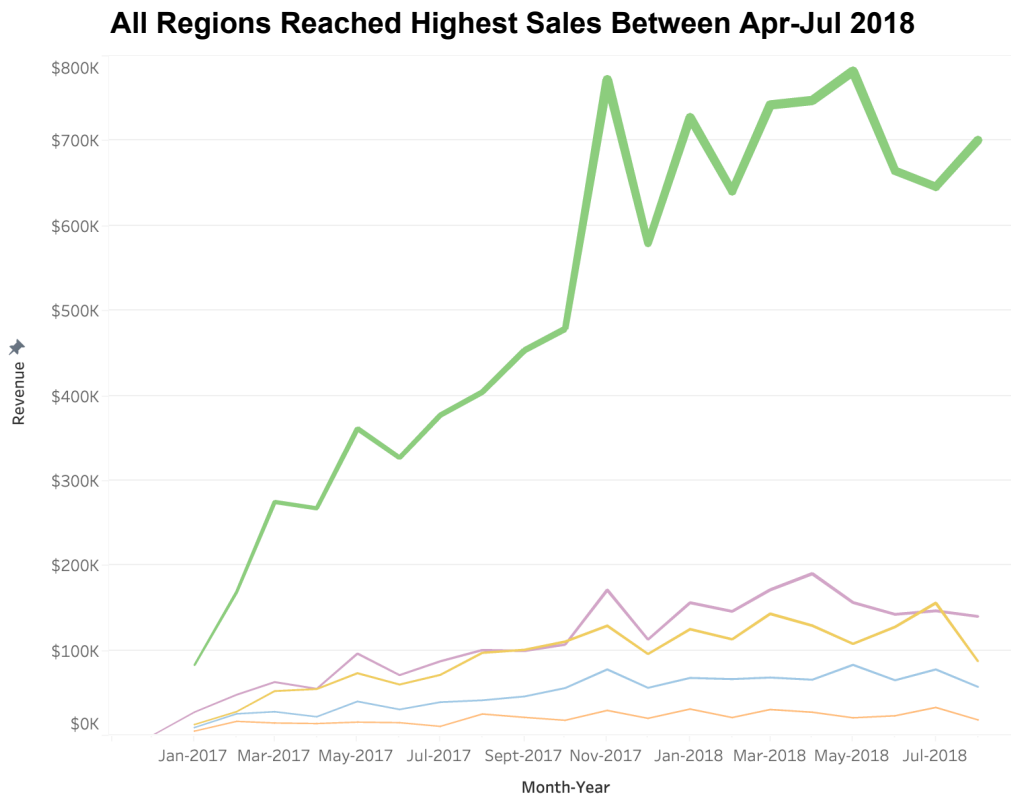
Delivery performance is the strongest predictor of customer review scores in the Olist dataset. The review vs delivery delay analysis demonstrates that each additional week of delay reduces average review scores by approximately 0.8 stars. States with persistently poor delivery performance are, by extension, generating structurally lower satisfaction scores that compound into lower repeat purchase rates and weaker word-of-mouth growth.

4.3 Seller Coverage vs Revenue Opportunity



The seller coverage scatter reveals a pronounced opportunity in the Northeast in Bahia state. This state sits alone in the top-left quadrant of the scatter: higher than average revenue and relatively low seller density, especially compared to neighbouring states in the region. This indicates demand outpacing supply. Targeted seller recruitment in BA could improve fulfilment economics (shorter last-mile distances, faster delivery) while capturing latent demand currently unmet or served by distant sellers.

4.4 Regional Revenue Trend

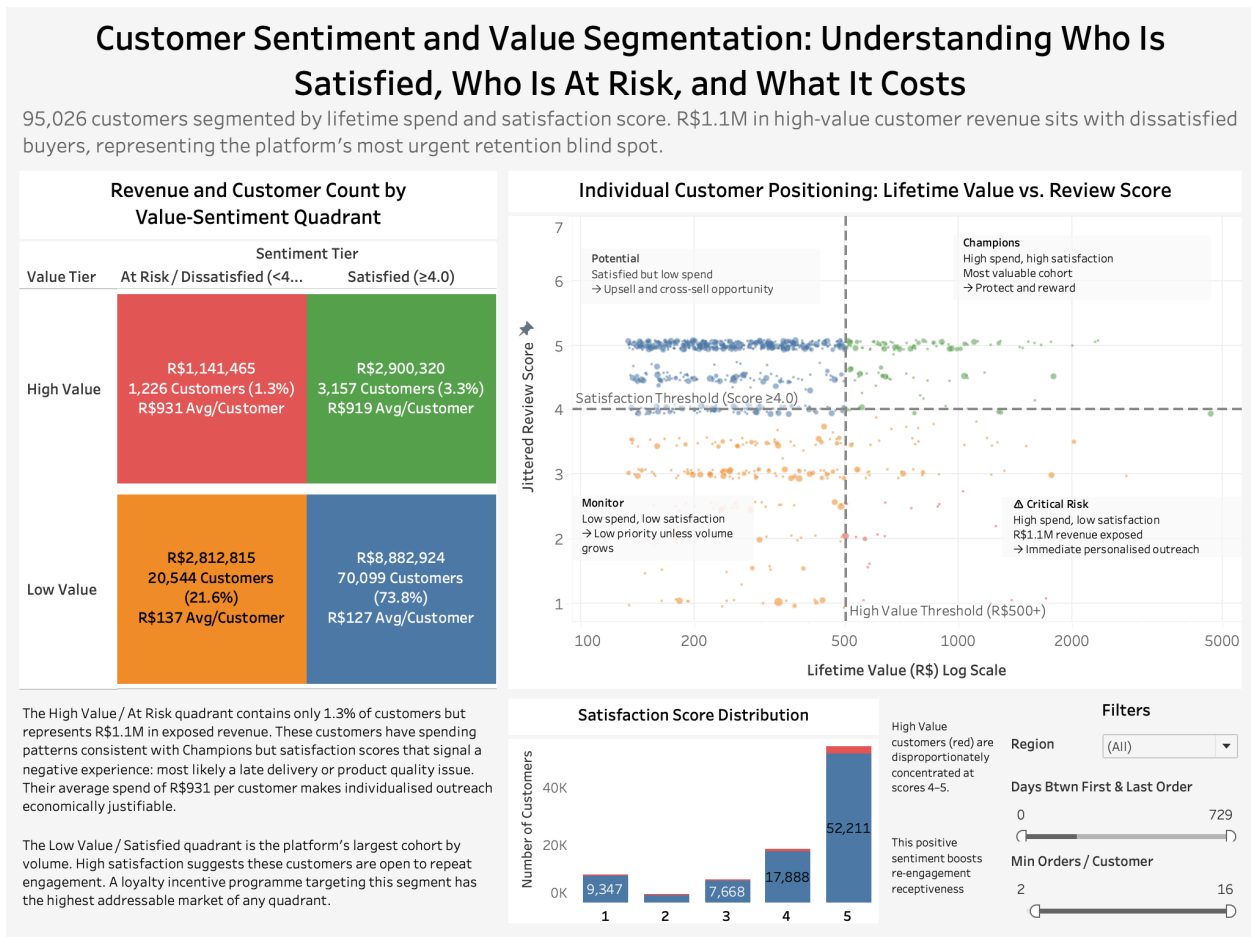


All five regions reached peak monthly revenue between April and July 2018, indicating that the platform-wide growth plateau observed in [Section 1.4](#) was a national rather than regionally isolated phenomenon. The Southeast maintained approximately 65% of total monthly revenue throughout 2018, with no meaningful convergence toward more balanced regional distribution. Geographic diversification will require active intervention.

Section 5: Customer Analysis

5.1 Customer Segmentation Framework

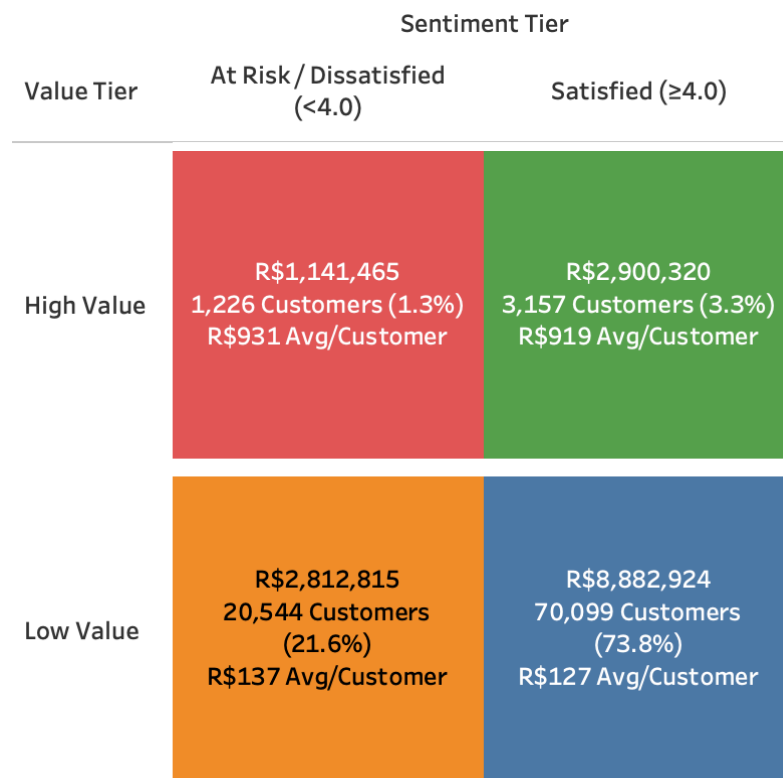
The customer analysis employs a two-dimensional segmentation framework combining a value tier (lifetime spend relative to the R\$500 high-value threshold) and a sentiment tier (average review score relative to the 4.0 satisfaction threshold). This produces four analytically distinct customer quadrants, each with different strategic implications.



5.2 Value-Sentiment Quadrant Analysis

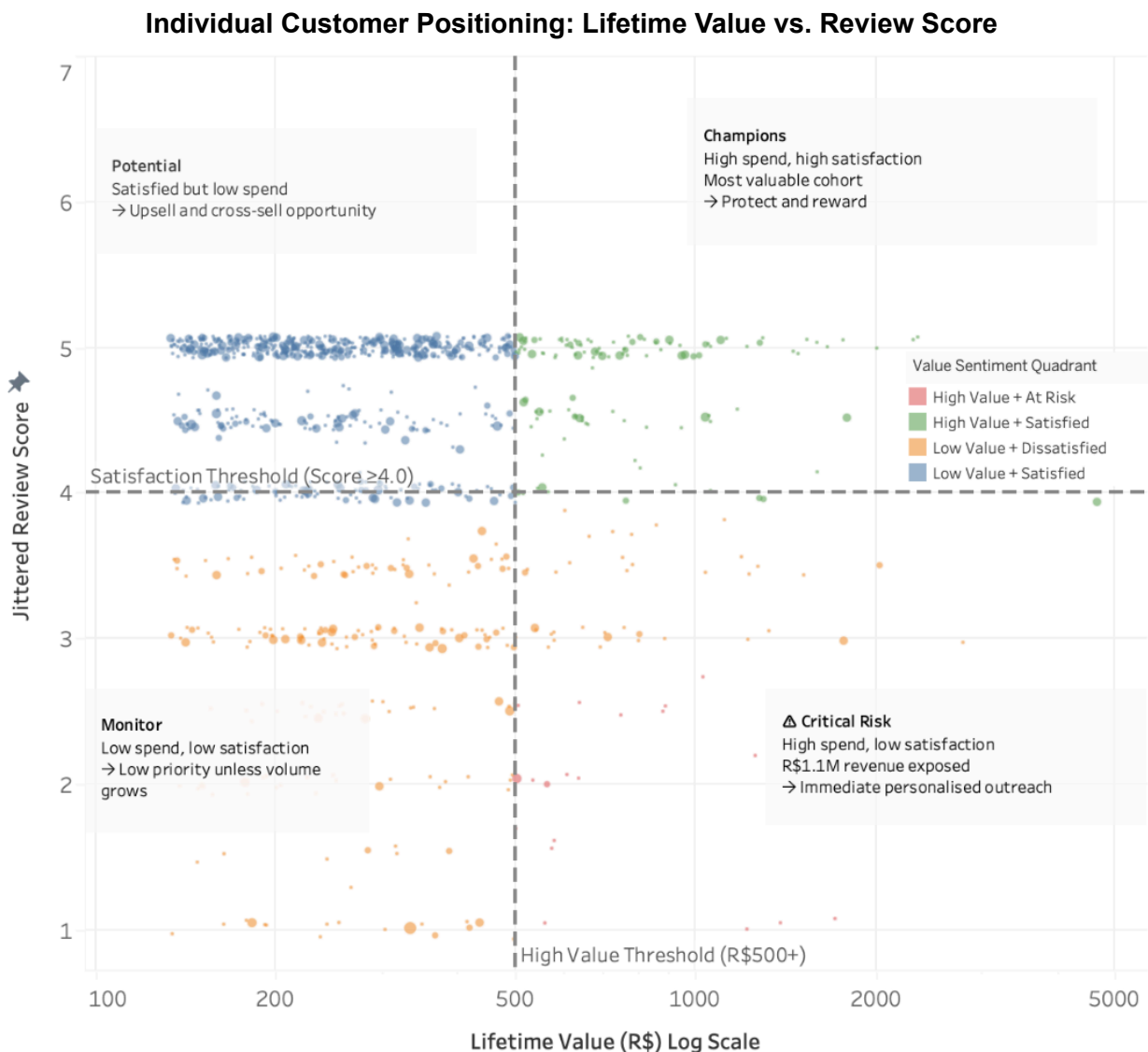
Quadrant	Customers	% of Total	Revenue	Avg per Customer	Strategic Priority
High Value + At Risk	1,226	1.3%	R\$1,141,465	R\$931	Immediate recovery
High Value + Satisfied	3,157	3.3%	R\$2,900,320	R\$919	Protect and grow
Low Value + At Risk	20,544	21.6%	R\$2,812,815	R\$137	Monitor
Low Value + Satisfied	70,099	73.8%	R\$8,882,924	R\$127	Loyalty nurture

Revenue and Customer Count by Value-Sentiment Quadrant



The difference in revenue per customer between the high and low value tiers is extreme: R\$931 compared with R\$137 (6.8x). This distinction justifies a higher cost of retention and recovery for high-value customers. Individualised outreach, personalised offers, and dedicated customer success contact are economically justifiable for the High Value + At Risk segment in a way that would be commercially unsustainable for the broader customer base.

5.3 Individual Customer Positioning



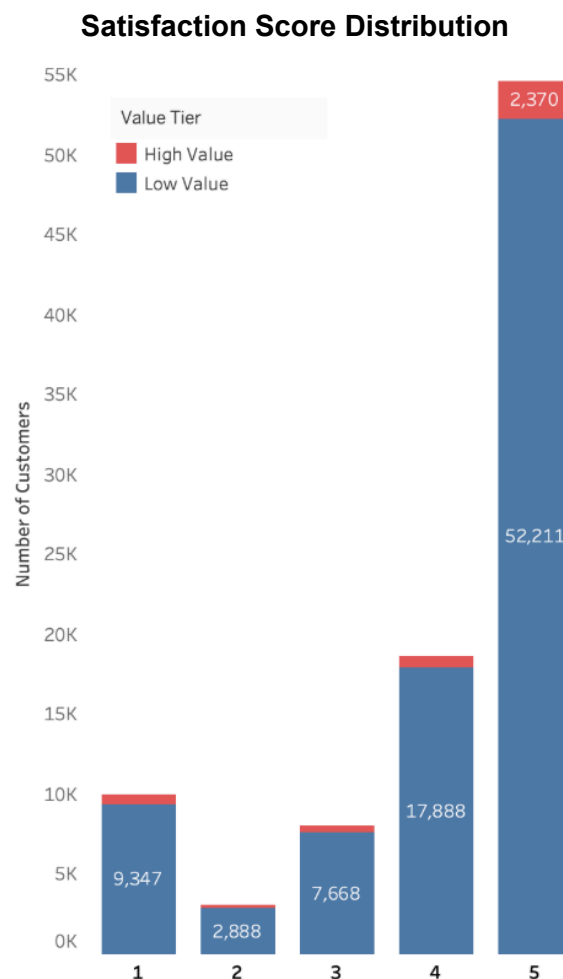
The scatter plot of 95,026 individual customers, each represented as a single dot positioned by lifetime value (x-axis, log scale) and average review score (y-axis), reveals the distribution within each quadrant. Three observations are particularly striking:

The High Value + At Risk cluster is not monolithic. High-value dissatisfied customers span a wide range of review scores from 1.0 to just below 4.0. Customers with scores of 3.5–3.9 are mildly dissatisfied and could be recoverable with a targeted win-back offer. But customers with scores between 1.0 and 2.0 had genuinely poor experiences. They will require a different recovery approach that acknowledges the specific failure before presenting any commercial offer.

The Champions cluster is well-defined. The top-right quadrant (high lifetime value, high satisfaction) contains a visible and distinct cluster of customers who have spent R\$1,000+ and consistently rate their experience at 4.5–5.0. These are the platform's brand advocates: customers most likely to respond to referral programmes and loyalty rewards.

Review scores cluster strongly at discrete values. The horizontal banding in the scatter (dense concentrations at scores of 1.0, 3.0, 4.0, 4.5, and 5.0) reflects that many customers have placed only a single order, giving them a review score equal to that one rating without averaging. This is an expected data characteristic of an emerging marketplace where repeat purchase rates are still developing.

5.4 Satisfaction Score Distribution

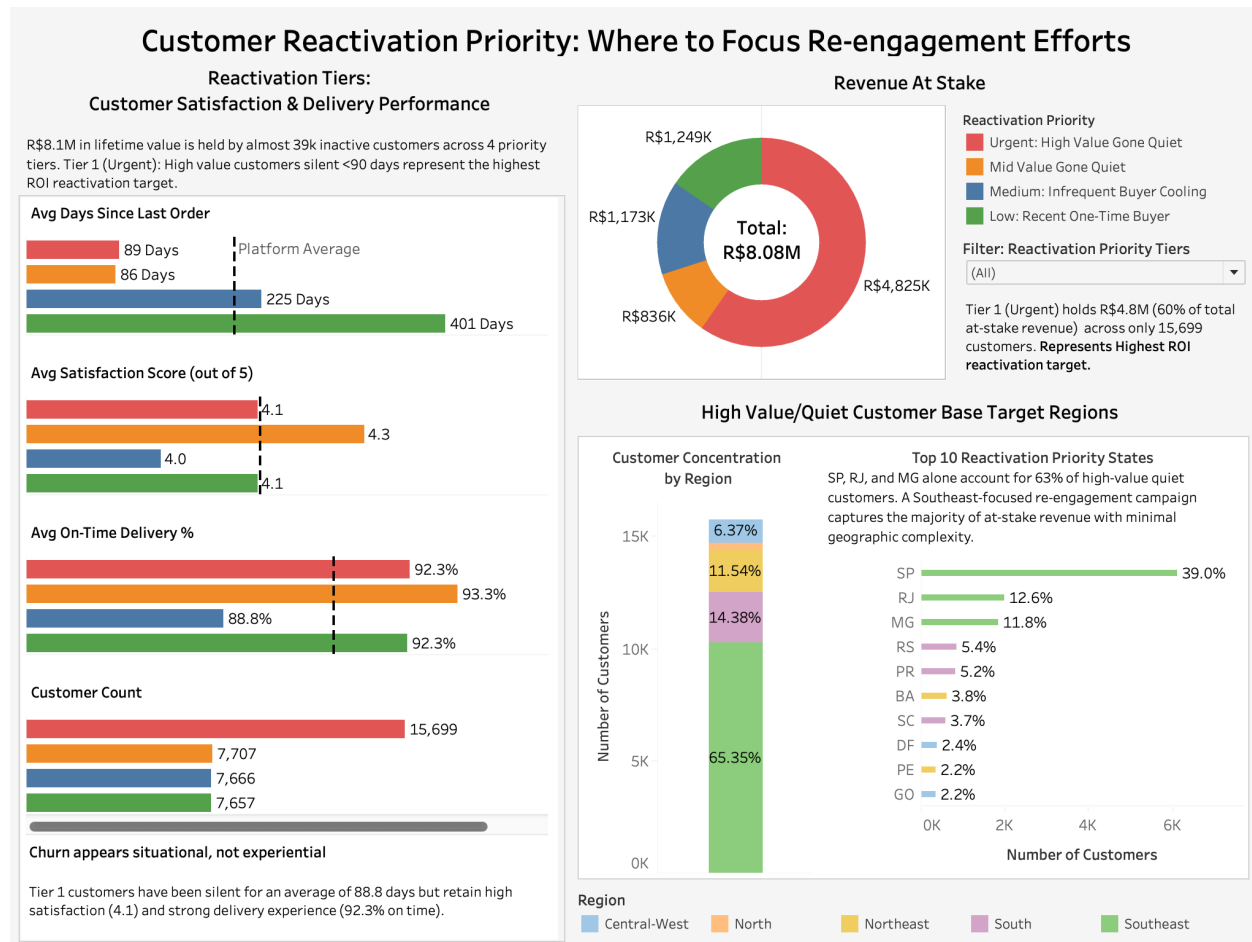


The satisfaction score distribution reveals a strongly positive platform sentiment: 73.9% of customers carry an average review score of 4.0 or higher. High Value customers are disproportionately concentrated at scores 4 and 5. They spend more and they are more satisfied than the broader customer base. This cohort's positive sentiment makes them highly

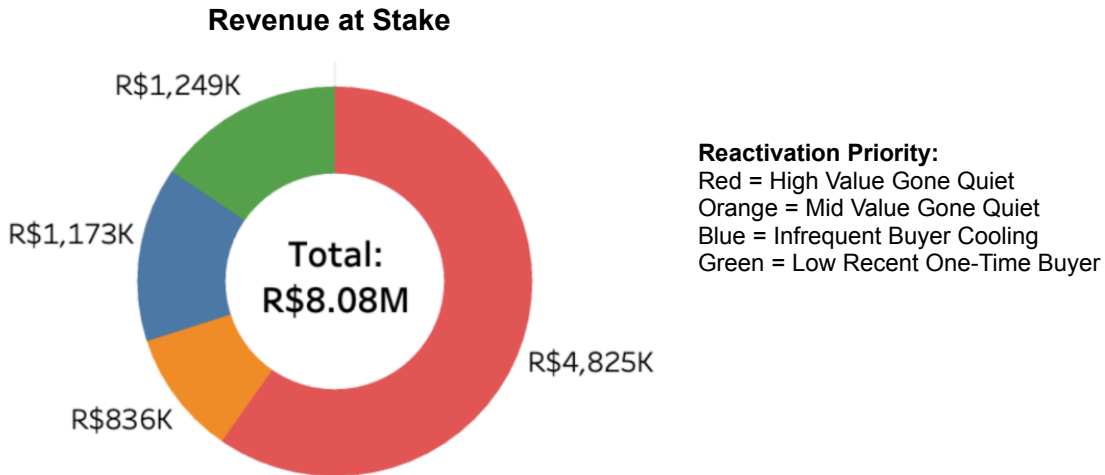
receptive to re-engagement, and their spending history makes the commercial case unambiguous. This is the highest-ROI re-engagement cohort on the platform.

The 9,347 customers at score bucket 1 (Very Dissatisfied) represent the platform's most at-risk segment. Understanding the root cause of their dissatisfaction (whether delivery failure, product quality, or post-purchase service) is a prerequisite for any recovery programme.

5.5 Customer Reactivation Priority Framework



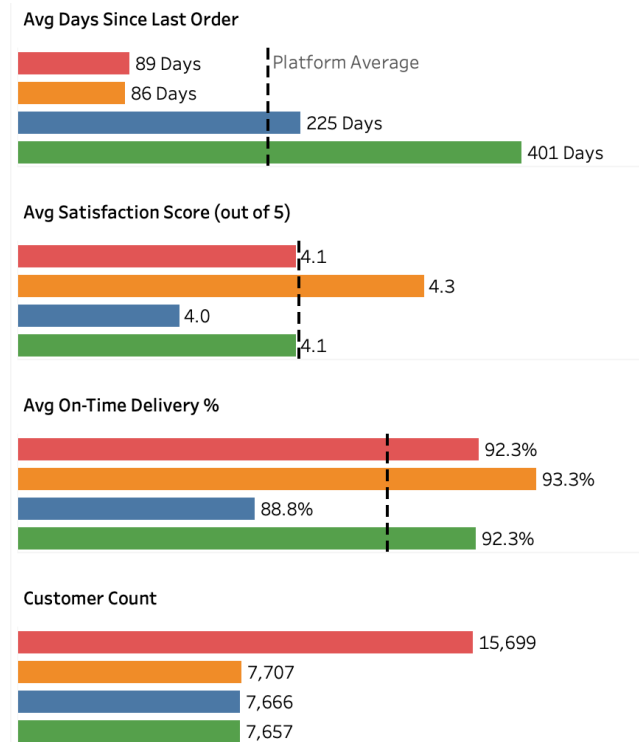
Building on the RFM (Recency, Frequency, Monetary) segmentation framework, inactive customers were assigned to four reactivation priority tiers based on their lifetime value and timing between repeat purchases. The total at-stake revenue across all four tiers is R\$8.08M.



Tier	Customers	Revenue at Stake	Avg Days Since Previous Order	Avg Satisfaction
1 — Urgent: High Value Gone Quiet	15,699	R\$4,825K	89 days	4.1
2 — Mid Value Gone Quiet	7,707	R\$836K	86 days	4.3
3 — Medium: Infrequent Buyer Cooling	7,666	R\$1,173K	225 days	4.0
4 — Low: Recent One-Time Buyer	7,657	R\$1,249K	401 days	4.1

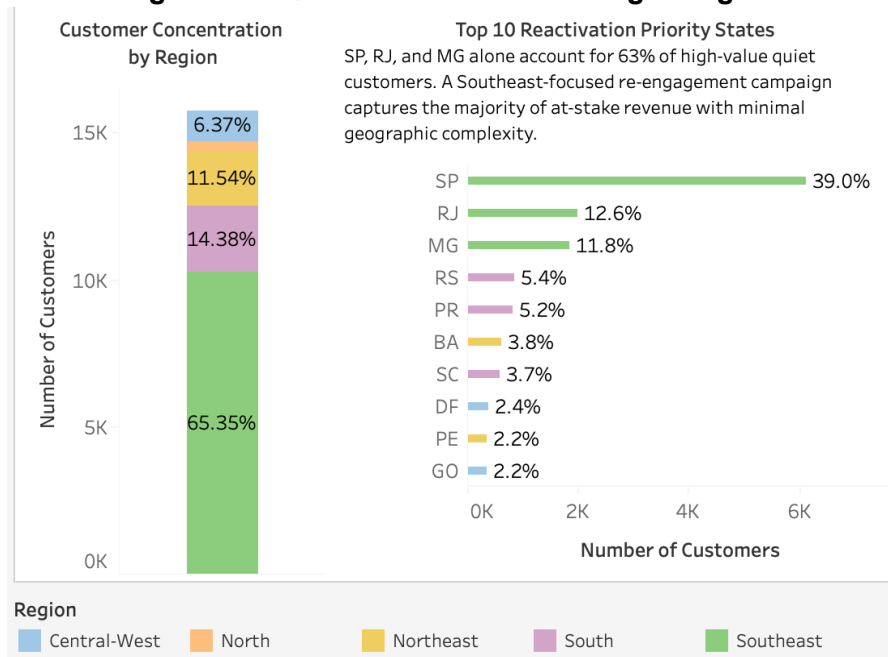
Tier 1 is the most commercially critical finding from this analysis. 15,699 customers hold R\$4.8M in at-stake lifetime value, which represents 60% of total at-risk revenue. They have been silent for an average of only 89 days. Critically, their average satisfaction score (4.1) and on-time delivery experience (92.3%) indicate that their inactivity is situational rather than experiential. They are not churning because of a bad experience, but because Olist has not given them a reason to return. This presents an opportunity: it means re-engagement campaigns face low friction and do not need to overcome negative sentiment before presenting an engaging offer.

Reactivation Tiers: Customer Satisfaction & Delivery Performance



Geographic targeting of reactivation campaigns can significantly improve efficiency. SP, RJ, and MG alone account for 63% of high-value quiet customers. A Southeast-focused re-engagement campaign captures the majority of at-stake revenue with minimal geographic complexity in campaign execution.

High Value/Quiet Customer Base Target Regions



Section 6: Methodology and Technical Architecture

6.1 Data Pipeline Architecture

This analysis was conducted using three-tier Medallion Architecture implemented on Databricks, processing the Olist dataset from raw CSV ingestion through to business-ready analytics tables.

Bronze (Raw Ingestion) → Silver (Cleaned & Validated) → Gold (Business-Ready Analytics)

Bronze Layer: All nine source CSV files were ingested into Delta tables using a dictionary-driven PySpark loop. An ingestion_timestamp audit column was added to each table. No transformations were applied: the bronze layer represents a faithful, unmodified copy of the source data.

Silver Layer: Each table was processed through a consistent workflow: schema inspection, NULL analysis, primary key validation, foreign key validation, business rule validation, transformation, and persistence to Delta format. Key transformations included:

- Custom PySpark UDF for Brazilian city name standardisation (accent removal, apostrophe handling, hyphen normalisation, title casing) applied across Customers, Geolocation, and Sellers tables.
- Review score field cleaned from a mixed-type field containing valid integer scores alongside erroneous timestamp strings that were entered in the wrong field.
- Geolocation table deduplicated from 1,000,163 rows to one representative coordinate per ZIP prefix using coordinate averaging after invalid coordinate removal.
- Order status consolidated and enriched with delivery performance flags.
- Product categories translated from Portuguese and consolidated from 73 granular values to 10 business-friendly macro-categories.
- Payment records cleaned: 11 invalid rows resolved (zero-value payments removed, missing instalment counts defaulted to 1)

All 6 critical foreign key relationships were validated using left anti joins. Payment totals were reconciled against order item totals with a 1-cent rounding tolerance.

Gold Layer: Silver tables were transformed into a star schema model using Spark SQL:

- 2 fact tables: gold_fact_order_items (item grain) and gold_fact_orders (order grain)
- 4 dimension tables: gold_dim_customers, gold_dim_products, gold_dim_sellers, gold_dim_dates
- 6 aggregate tables: daily sales, monthly sales, product performance, top sellers, customer lifespan, regional sales

Each gold table was followed by a dedicated validation cell verifying row counts, primary key uniqueness, range constraints, and business rule compliance.

6.2 Key Analytical SQL Techniques

Gold layer tables served as the exclusive data source for all SQL queries across the item, product, seller, regional, and customer dimensions of this analysis. This ensured every metric was derived from validated, transformation-complete data rather than raw source tables. All visualisations were built and published using Tableau Public.

Where queries joined fact tables across different grains — for example, combining order-level revenue from `gold_fact_orders` with item-level seller data from `gold_fact_order_items` — aggregations were deliberately separated into independent CTEs before joining to prevent fan-out inflation of revenue and delivery metrics.

6.3 Data Quality Findings

The analysis identified and resolved several noteworthy data quality issues in the source data:

Review score field corruption: The `review_score` column in the orders review table contained a mixture of valid integer scores (1–5) and timestamp strings entered incorrectly into the wrong field. A cleaned column was derived by extracting only values parseable as integers in the 1–5 range, with malformed rows removed from the silver table.

Orders without items or payments: 776 orders existed in the orders table with no corresponding records in order items or payments, representing orders cancelled before any transaction processing. These were intentionally excluded from `gold_fact_orders` via inner join and explicitly documented in the validation cell with the expected row difference noted.

Payment reconciliation discrepancies: 11 payment records had either zero payment values or missing instalment counts. Zero-value payments were quarantined and removed. Positive-value payments with missing instalment counts were defaulted to 1 (single payment assumption). A payment reconciliation check confirmed that the sum of payment values matched the sum of order item totals within a 1-cent tolerance across all included orders.

Cancelled orders with delivery dates: 6 cancelled orders retained delivery dates. This is a real-world edge case where packages already in transit were delivered despite cancellation status being recorded. These rows were retained with a dedicated `delivery_performance` flag value of Canceled — Delivered.

Section 7: Strategic Recommendations

7.1 Priority Matrix

The findings from this analysis can be organised into a 2×2 priority matrix based on commercial impact and implementation complexity:

High Impact, Low Complexity — Pursue Immediately:

- Launch a targeted re-engagement campaign for Tier 1 reactivation customers. This cohort of 15,699 customers holds R\$4.8M at stake and has been silent for an average of only 89 days, with high satisfaction scores that create a low-friction outreach environment
- Initiate SLA improvement conversations with High Revenue / Poor Delivery sellers (321 sellers, 48% of GMV, 13.9 avg delivery days)
- Personalised recovery outreach for High Value + At Risk customers (1,226 customers, R\$1.1M at stake, R\$931 average spend per customer)

High Impact, Higher Complexity — Plan and Execute:

- Develop a formal seller retention programme for the top 136 sellers (4.4% of sellers, 50% of GMV)
- Implement a regional seller recruitment programme that targets the Northeast and North regions (lower delivery performance, untapped demand)
- Review and delist or reprice 904 pipeline risk products (freight ratio $\geq 30\%$ with sustained demand decline)

Medium Impact, Low Complexity — Quick Wins:

- Introduce a loyalty incentive programme for Low Value + Satisfied segment (70,099 customers, highest addressable market for volume-driven repeat purchase growth)
- Provide commercial support (marketing co-investment, catalogue expansion) to Low Revenue / High Satisfaction sellers. Operational excellence is already present; only commercial acceleration is required
- Establish minimum performance thresholds for Underperformer tier sellers with defined review period

Longer Term — Strategic Investment:

- Demand generation programmes in the Northeast markets in BA, which has seller coverage gaps relative to neighbouring states
- Category diversification strategy to reduce dependence on top 3 revenue categories (currently approximately 50% of total GMV)
- Premium shipping tier for high-value items to reduce long-tail delivery outliers

7.2 Revenue Impact Modelling

The strategic recommendations above can be quantified in terms of recoverable revenue:

Initiative	Revenue at Stake	Recovery Assumption*	Recoverable Revenue*
Tier 1 reactivation campaign	R\$4,825K	20% conversion	R\$965K
High Value + At Risk recovery	R\$1,141K	35% recovery	R\$399K
Pipeline risk product margin improvement	R\$541K	40% margin recovery	R\$216K
Low Value + Satisfied loyalty programme	R\$8,883K	15% uplift	R\$1,332K
Total recoverable opportunity			~R\$2.9M
*Recovery assumptions are illustrative benchmarks based on general e-commerce industry ranges and are not derived from Olist platform data. Actual recoverable revenue will depend on campaign execution, pricing decisions, and market conditions. These figures are presented to demonstrate the order-of-magnitude opportunity rather than as precise forecasts.			

These assumptions are conservative. A well-executed Tier 1 reactivation campaign targeting customers with high satisfaction scores and relatively recent last orders could realistically exceed 20% conversion given the low friction environment. Separately, the Low Value + Satisfied segment's demonstrated platform trust supports the loyalty programme uplift assumption.

Section 8: Conclusions

This analysis demonstrates that the Olist platform possesses strong operational fundamentals: a 97.8% delivery success rate, a 4.1 average review score, and consistent revenue growth through 2017. However, there are also structural vulnerabilities that will require deliberate strategic intervention to address. The central theme across all five analytical dimensions is concentration: revenue concentrated in a small seller cohort, geographic revenue concentrated in the Southeast, customer satisfaction risk concentrated in a specific high-value dissatisfied segment, freight margin risk concentrated in a subset of low-price categories, and category revenue concentrated in three product families. In each case concentration reflects the natural dynamics of a growing marketplace, but it creates specific, quantifiable risks that are addressable with the right analytical infrastructure in place.

The most immediately actionable finding is the convergence of high commercial value and low recovery friction in the Tier 1 reactivation segment. 15,699 customers holding R\$4.8M in lifetime value have been quiet for fewer than 90 days on average, with satisfaction and delivery scores that indicate their inactivity is situational rather than experiential. This is not a churned customer problem. It is an engagement gap, which is far more malleable than failures that are rooted in a genuinely poor experience. The distinction matters strategically: campaigns targeting this cohort do not need to overcome negative sentiment before presenting a commercial offer, which materially improves the economics of outreach.

What This Pipeline Enables Next

The Medallion Architecture pipeline built for this project, nine validated silver tables feeding a twelve-table gold layer star schema, is not a one-time analytical exercise. It is a reusable infrastructure that can support a more sophisticated analytical programme as the platform scales. Three high-value extensions are worth prioritising.

Predictive churn modelling: The RFM segmentation in this report is descriptive. It identifies which customers have already gone quiet. A logical next step would be to build a predictive model that scores active customers on their probability of lapsing before they do, using the recency, frequency, and satisfaction signals already present in `gold_agg_customer_lifespan`. A logistic regression, random forest, or gradient-boosted tree model trained on historical lapse patterns could produce a monthly churn risk score per customer, enabling proactive intervention rather than re-engagement. The gold layer's clean, validated data makes this a pipeline-ready input rather than a data preparation project.

A/B testing framework for reactivation campaigns: The Tier 1 reactivation opportunity identified in Section 5.5 is compelling, but the recovery assumptions in Section 7.2 are illustrative rather than empirically grounded. Instrumenting a controlled experiment would replace those assumptions with platform-specific conversion data. Varying offer types (discount vs loyalty points vs free shipping), communication channels (email vs push notification), and timing (30-day vs 60-day vs 90-day post-lapse trigger) are key conditions to test.

Regional demand forecasting: The regional analysis in Section 4 established the current state of geographic revenue distribution but did not model where demand is likely to grow.

Incorporating external data sources such as population growth projections, internet penetration rates by state, competitor marketplace expansion announcements would enable a demand forecasting model that prioritises seller recruitment geographically based on projected rather than observed revenue gaps.

The most important characteristic of a data-mature organisation is not the sophistication of its models but the quality of its questions and the reliability of the infrastructure that answers them. This project has built both. The questions are commercially grounded and analytically tractable: which customers are worth recovering, which sellers are worth protecting, which products are worth stocking, and which regions are worth investing in. The pipeline that answers them is validated, documented, and extensible. The analysis proposals described above are not aspirational additions requiring new infrastructure: they are the natural next iteration of this work.

Appendix A: Data Sources and Pipeline Summary

The following table summarises the technology, table count, and purpose of each pipeline layer.

Layer	Technology	Tables/Visualisations	Purpose
Bronze	Databricks, PySpark	9 tables	Raw CSV ingestion
Silver	Databricks, PySpark	9 tables + 1 quarantine	Cleaning and validation
Gold	Databricks, Spark SQL	12 tables (2 fact, 4 dim, 6 agg)	Business analytics
Visualisation	Tableau Public	7 dashboards	Reporting and insight delivery

Source Dataset: Brazilian E-commerce Public Dataset by Olist

Dataset URL: <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

Observation Period: 2016–2018

License: CC BY-NC-SA 4.0

Appendix B: Gold Layer Table Reference

The following table describes the grain and primary analytical use of each gold layer table.

Table	Grain	Primary Use
gold_fact_order_items	One row per order line item	Item-level revenue and delivery analysis
gold_fact_orders	One row per order	Order-level KPIs, payment reconciliation
gold_dim_customers	One row per customer_id	Geographic and identity context
gold_dim_products	One row per product	Catalogue and freight analysis
gold_dim_sellers	One row per seller	Seller performance and geography
gold_dim_dates	One row per calendar date	Time intelligence across all fact tables
gold_agg_daily_sales	One row per day	Daily revenue and delivery monitoring
gold_agg_monthly_sales	One row per month	Trend analysis with MoM metrics
gold_agg_product_performance	One row per product	Product revenue and review analytics
gold_agg_top_sellers	One row per seller	Pareto and concentration analysis
gold_agg_customer_lifespan	One row per customer_unique_id	LTV, RFM, and lifecycle segmentation
gold_agg_regional_sales	One row per region/state	Geographic performance analysis